Cisco Networking Academy
Mind Wide Open

Chapter 8: Example Subnetting IP networks

Introduction to Networks v5.1

Review

Classless inter-domain routing (**CIDR**)

Metode yang digunakan ISP untuk alokasikan sejumlah alamat pada suatu perusahaan

Class C

Binary	Decimal	Singkatan	
10000000	= 128	/25	(tidak vailid untuk ujian cisco)
11000000	= 192	/26	
11100000	= 224	/27	
11110000	= 240	/28	
11111000	= 248	/29	
11111100	= 252	/30	
11111110	= 254	/31	(tidak vailid)

Example 1:
255.255.255.192
192 = 1 1 0 0 0 0 0 0

- **Bit 1 mewakili bit bit subnet, Bit 0 mewakili bit bit host yang tersedia setiap subnet.**
- **192 memberikan 2 Bit untuk sebnet subnet, 6 Bit mendefenisikan host host pada masing masing subnet**
- **Karena bit bit subnet tidak boleh semuanya off ((bernilai 0) atau on (bernilai 1 semua) pada saat bersamaan**

Maka hanya 2 subnet yang vailid :
01000000 = 64 (semua bit host off)
10000000 = 128 (semua bit host off)

Alamat dari host-host yang vailid akan didevenisikan sebagai nomor-nomor diantara subnet-subnet tersebut dikurangi dengan dua nomor :

a. **Nomor yang semua bit host bernilai 0 (off)**
b. **Nomor yang semua bit host bernilai 1 (on)**

Untuk menentukan host-host ?

1. **Menemukan subnet dengan membuat semua bit host off**
2. **Kemudian membuat semua bit host on untuk mencari alamat broadcast untuk subnet tersebut**
3. **Host yang vailid harus berada diantara kedua nomor atau alamat**

01000000 = 64 (semua bit host off)

Subnet 64

Subnet	Host		Artinya
01	000000	= 64	Alamat Subnet
01	000001	= 65	Host 1 yang vailid
01	111110	= 126	Host terakhir yang vailid
01	111111	= 127	Alamat Broadcast

64 Merupakan **Binary** **Desimal**
Baris pertama, subnet (01) + Host (000000) = 01 000000 = 64

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Review

Subnet 128

Subnet	Host		Artinya
10	000000	= 128	Alamat Subnet
10	000001	= 129	Host 1 yang vailid
10	111110	= 190	Host terakhir yang vailid
10	111111	= 191	Alamat Broadcast

127 Merupakan **Binary** Desimal

Baris pertama, subnet (10) + Host (000000) = 01 000000 = 191

Subnetting pada ip address class c

Berikutnya kita akan mencoba melakukan subnetting untuk IP address class B. Pertama, subnet mask yang bisa digunakan untuk subnetting class B adalah seperti dibawah. Sengaja saya pisahkan jadi dua, blok sebelah kiri dan kanan karena masing-masing berbeda teknik terutama untuk oktet yang "dimainkan" berdasarkan blok subnetnya. CIDR /17 sampai /24 caranya sama persis dengan subnetting Class C, hanya blok subnetnya kita masukkan langsung ke oktet ketiga, bukan seperti Class C yang "dimainkan" di oktet keempat. Sedangkan CIDR /25 sampai /30 (kelipatan) blok subnet kita "mainkan" di oktet keempat, tapi setelah selesai oktet ketiga berjalan maju (coeunter) dari 0, 1, 2, 3, dst.

Subnetting pada ip address class a

Konsepnya semua sama saja. Perbedaannya adalah di **OKTET** mana kita mainkan blok subnet. Kalau Class C di oktet ke 4 (terakhir), kelas B di Oktet 3 dan 4 (2 oktet terakhir), kalau Class A di oktet 2, 3 dan 4 (3 oktet terakhir). Kemudian subnet mask yang bisa digunakan untuk subnetting class A adalah semua subnet mask dari CIDR /8 sampai /30.

Cara Cepat Subnetting

Class C
255.255.255.192
Melakukan subnet terhadap alamat network 192.168.10.0 dan subnet mask 255.255.255.192 (/26)

Alamat Net : 192.168.10.0
Subnet Mask : 255.255.255.192

Berapa Banyak Subnet ?
192 mempunyai 2 bit 1 atau 2 bit on (11000000).

$$2^n - 2 = 2^2 - 2 = 2$$

Berapa banyak host per subnet ?
192 mempunyai 6 bit off (11 000000).

$$2^n - 2 = 2^6 - 2 = 62 \text{ host}$$

Berapa jumlah subnet subnet yang vailid ?

$$256 - 192 = 64$$

- Subnet pertama dan base number atau block size
- Tambahkan Block size sampai mencapai mencapai subnet mask

64 + 64 = 128 dan 128 + 64 = 192

➔ Tidak vailid; subnet mask (semua bit subnet on
➔ Dua subnet yang vailid adalah 64 dan 128

Alamat broadcast untuk setiap subnet ?
Nomor yang berada tepat sebelum subnet berikutnya, yang mempunyai semua bit on (bernilai 1)

Host host yang vailid ?

Subnet 64

Subnet1	Subnet2	Artinya
64	128	Subnet (langkah 1)
65	129	Host 1 yang vailid
126	190	Host terakhir yang vailid
127	191	Alamat Broadcast (langkah 2)

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Review

Class C
255.255.255.224 (/27)
Melakukan subnet terhadap alamat network 192.168.10.0 dan subnet mask 255.255.255.224 (/27)
Alamat Net : 192.168.10.0
Subnet Mask : 255.255.255.224
Berapa Banyak Subnet ?
224 mempunyai 3 bit 1 atau 3 bit on (111 00000).

$$2^n-2 = 2^3-2 = 6$$

Berapa banyak host per subnet ?
224 mempunyai 5 bit off (111 00000).

$$2^n-2 = 2^5-2 = 30 \text{ host}$$

Berapa jumlah subnet subnet yang vailid ?

$$256 - 224 = 32$$

- Subnet pertama dan base number atau block size
- Tambahkan Block size sampai mencapai mencapai subnet mask

32 + 32 = 64 ; 64 + 32 = 96 ; 96 + 32 = 128 ; 128 + 32 = 160 ; 160 + 32 = 192 ; 192 + 32 = 224

➔ Tidak vailid; subnet mask (semua bit subnet on
➔ Enam subnet yang vailid

Alamat broadcast untuk setiap subnet ?
Nomor yang berada tepat sebelum subnet berikutnya, yang mempunyai semua bit on (bernilai 1)
Host host yang vailid ?

Class C
255.255.255.248 (/29)
Melakukan subnet terhadap alamat network 192.168.10.0 dan subnet mask 255.255.255.248

Berapa Banyak Subnet ?
248 mempunyai 5 bit 1 atau 5 bit on (11111 000).

$$2^n-2 = 2^5-2 = 30$$

Berapa banyak host per subnet ?
248 mempunyai 5 bit off (11111 000).

$$2^n-2 = 2^5-2 = 6 \text{ host}$$

Berapa jumlah subnet subnet yang vailid ?

$$256 - 248 = 8$$

- Subnet pertama dan base number atau block size
- Tambahkan Block size sampai mencapai mencapai subnet mask

8, 16, 32, ...224,232,240

➔ Tidak vailid; subnet mask (semua bit subnet on
➔ Enam subnet yang vailid

Alamat broadcast untuk setiap subnet ?
Nomor yang berada tepat sebelum subnet berikutnya, yang mempunyai semua bit on (bernilai 1)
Host host yang vailid ?

8	**16**	**24**	**.....**	**224**	**232**	**240**	**Subnet**
9	**17**	**25**	**.....**	**225**	**233**	**241**	**Host 1 yang vailid**
14	**21**	**20**	**.....**	**230**	**238**	**246**	**Host terakhir yang vailid**
15	**23**	**31**	**.....**	**231**	**239**	**247**	**Alamat Broadcast (langkah 2)**

S= subnet

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Review

Class C
255.255.255.252 (/30)
Melakukan subnet terhadap alamat network 192.168.10.0 dan subnet mask 255.255.255.252 (/30)
Alamat Net : 192.168.10.0
Subnet Mask : 255.255.255.252
Berapa Banyak Subnet ?
252 mempunyai 6 bit 1 atau 6 bit on (111111 00).

$$2^n-2 = 2^6-2 = 62$$

Berapa banyak host per subnet ?
224 mempunyai 5 bit off (111 00000).

$$2^n-2 = 2^2-2 = 2 \text{ host}$$

Berapa jumlah subnet subnet yang vailid ?

$$256 - 252 = 4$$

- Subnet pertama dan base number atau block size
- Tambahkan Block size sampai mencapai mencapai subnet mask
  4; 8; 12 . . . . . .; 248
- ➔ Tidak vailid; subnet mask (semua bit subnet on
- ➔ Enam subnet yang vailid

Alamat broadcast untuk setiap subnet ?
Nomor yang berada tepat sebelum subnet berikutnya, yang mempunyai semua bit on (bernilai 1)
Host host yang vailid ?

4	8	12 248	Subnet (langkah 1)
5	9	13 249	Host 1 yang vailid
6	10	14 250	Host terakhir yang vailid
7	11	15 251	Alamat Broadcast (langkah 2)

VLSM

Case : **Seorang Network engineer, menemukan PT. X, dengan lusinan LINK WAN yang terhubung dengan Kantor pusat anda. Saat ini network anda adalah sebuah network yang classful, yang berarti Semua host dan interface router memiliki subnet mask yang sama pada setiap interface.**
Classless routing, dimana kiat dapat memilih mask mask yang berbeda dengan ukuran yang berbeda. Akan tetapianda tidak tau apa yang harus digunakan utuk link WAN point -to -point anda.

Apakah 255.255.255.0
Mask yang dapat membantu dalam situasi ini ? WAN

Melakukan subnet terhadap alamat network 192.168.10.0 dan subnet mask 255.255.255.248

- Jika menggunakan mask 255.255.255.0, maka setiap network memiliki 254 host
- Keperluan link WAN point -to -point , hanya memerlukan dua buah alamat host saja untuk setiap subnet !
- Akan menyia nyiakan 252 host pada setiap subnet

- ➢ Jika menggunakan mask 255.255.255.252, maka setiap network memiliki 2 host
- ➢ Tida menyia nyiakan alamat IP yang berharga

4	8	12 240	244	248	Subnet (langkah 1)
5	9	13 241	245	249	Host 1 yang vailid
6	10	14 242	246	250	Host terakhir yang vailid
7	11	15 243	247	251	Alamat Broadcast (langkah 2)

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Review

Subnet di Kepala .....!!!!!!!

Class C
Alamat Node : 192.168.10.33
Subnet Mask : 255.255.255.224

Good

Tentukan Subnet dan Alamat Broadcast dari alamat alamat IP diatas ?

a. Subnet subnet apakah yang vailid ?
256 - 224 = 32 ; 32 + 32 = 64
b. Alamat Node 192.168.10.33 **; berada diantara dua subnet**
c. Yang pasti, merupakan bagian dari subnet Subnet 192.168.10.32
d. Subnet berikutnya 64
e. Alamat Broadcast selalu nomor yang berbeda, persis sebelum subnet sebelumnya
f. Range Host yang vailid 33 – 62

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Review : Class B

Jauh Lebih banyak subnet dari pada Class C

172.16.0.0 Ada 16 bit tersedia untuk alamat HOST
Kita dapat menggunakan sampai 14 bit untuk subnetting
Karena kita harus meninggalkan 2 bit untuk pengalamatan host

255.255.128.0	(/17)	255.255.255.128	(/24)
255.255.192.0	(/18)	255.255.225.128	(/25)
255.255.224.0	(/19)	255.255.255.192	(/26)
255.255.240.0	(/20)	255.255.255.224	(/27)
255.255.248.0	(/21)	255.255.255.240	(/28)
255.255.252.0	(/22)	255.255.255.248	(/29)
255.255.254.0	(/23)	255.255.255.252	(/30)

Pola yang digunakan yaitu :
128,192,224,240,248,252, dan 254
128 10000000 : 1 bit subnet
192 11000000 : 2 bit subnet

Good

Class B

Alamat subnet	16 0	32.0
Alamat Broadcast	16.255	32.255

Tambahkan alamat host yang vailid diantara nomot nomor tersebut

Class B
255.255.192.0 (/18)
Melakukan subnet terhadap alamat network 172.16.0.0 dan subnet mask 255.255.192.0

Berapa Banyak Subnet ?
192.0 mempunyai 18 bit 1 atau 18 bit on
(11000000 00000000).

$$2^n-2 = 2^2-2 = 2$$

Berapa banyak host per subnet ?
224 mempunyai 5 bit off (111 00000).

$$2^n-2 = 2^{14}-2 = 16.382$$ (6 bit di oktet ke 3 dan 8 bit si oktet ke 4

Berapa jumlah subnet yang vailid ?

$$256 - 192 = 64$$

- Subnet pertama dan base number atau block size
- Tambahkan Block size sampai mencapai mencapai subnet mask

64 + 64 = 128

Alamat broadcast untuk setiap subnet ?
Nomor yang berada tepat sebelum subnet berikutnya, yang mempunyai semua bit on (bernilai 1)

Host host yang vailid ?

Alamat subnet	64 . 0	128 . 0
Host Pertama	64 . 1	128 . 1
Host terakhir	127 . 254	191 . 254
Alamat Broadcast	127 . 255	191 . 255

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Class B
255.255.240.0 (/20)
Melakukan subnet terhadap alamat network 172.16.0.0 dan subnet mask 255.255.240.0

Berapa Banyak Subnet ?

(11110000 00000000).

$2^n-2 = 2^4-2 = 14$

Berapa banyak host per subnet ?

$2^n-2 = 2^{12}-2 = 4094$

Berapa jumlah subnet yang vailid ?

256 - 240 = 16

16; 16 + 16 = 32; 32 + 16 = 48 224

Ini Sama dengan Class C

Alamat broadcast untuk setiap subnet ?

Host host yang vailid ?

Alamat subnet	16 . 0	32 . 0 . . . .
Host Pertama	16 . 1	32 . 1 . . . .
Host terakhir	31 . 254	47 . 254 . . .
Alamat Broadcast	31 . 255	47 . 255 . . .

Class B
255.255.254.0 (/23)
Melakukan subnet terhadap alamat network 172.16.0.0 dan subnet mask 255.255.254.0

Berapa Banyak Subnet ?

(11111110 00000000).

$2^n-2 = 2^7-2 = 126$

Berapa banyak host per subnet ?

$2^n-2 = 2^9-2 = 510$

Berapa jumlah subnet yang vailid ?

256 - 254 = 2

2; 2 + 2 = 4; 4 + 2 = 6 224

Alamat broadcast untuk setiap subnet ?

Host host yang vailid ?

Alamat subnet	2 . 0	4 . 0 . . . .
Host Pertama	2 . 1	4 . 1 . . . .
Host terakhir	3 . 254	5 . 254 . . . .
Alamat Broadcast	3 . 255	5 . 255 . . . .

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Review

Subnet di Kepala .....!!!!!!!

Class B

Buat Pening

Class B

Alamat Node : 172.16.10.33

Subnet Mask : 255.255.255.224

Tentukan Subnet dan Alamat Broadcast dari alamat alamat IP diatas ?

a. Subnet subnet apakah yang vailid ?
   256 - 224 = 32 ; 32 + 32 = 64 ; Cek : 33 di antara 32 dan 64
b. Ingat oktet ketiga merupakan bagian dari subnet
c. Jadi jawabanya adalah subnet 10.32 dan lengkapnya 172.16.10.32
d. Alamat Broadcast adalah subnet 10.33 dan lengkapnya 172.16.10.33
e. Karena 10.64 adalah Subnet berikutnya
f. **Apakah Subnet** dan **Alamat Broadcast** dari alamat 172.16.90.66 **Subnet Mask : 255.255.255.192 ?**
   256 - 192 = 64
   64 + 64 = 128.

g. Subnet 172.16.90.64 ; Broadcast 172.16.90.127, karena 50.128 adalah subnet berikutnya

h. **Apakah Subnet** dan **Broadcast** dari alamat 172.16.50.97 **Subnet Mask : 255.255.255.224 ?**
   256 - 224 = 32
   32;64;96;128.

i. Subnet 172.16.50.96 ; Broadcast 172.16.50.127, karena 50.128 adalah subnet berikutnya

j. **Apakah Subnet** dan **Broadcast** dari alamat 172.16.10.10 **Subnet Mask : 255.255.255.192 ?**
   256 - 192 = 64

k. Alamat ini berbeda di Subnet 172.16.10.0 ; Broadcast 172.16.10.63

l. **Apakah Subnet** dan **Broadcast** dari alamat 172.16.10.10 **Subnet Mask : 255.255.255.252 ?**
   256 - 252 = 4

m. Subnet 172.16.50.96 ; Broadcast 172.16.10.8, Broadcast 172.16.10.11

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Review : Class A

Jauh Lebih banyak subnet dari pada Class C

172.16.0.0 Ada 16 bit tersedia untuk alamat HOST
Kita dapat menggunakan sampai 14 bit untuk subnetting
Karena kita harus meninggalkan 2 bit untuk pengalamatan host

Subnet Mask	Nilai CIDR
255.128.0.0	/9
255.192.0.0	/10
255.224.0.0	/11
255.240.0.0	/12
255.248.0.0	/13
255.252.0.0	/14
255.254.0.0	/15
255.255.0.0	/16
255.255.128.0	/17
255.255.192.0	/18
255.255.224.0	/19

Subnet Mask	Nilai CIDR
255.255.240.0	/20
255.255.248.0	/21
255.255.252.0	/22
255.255.254.0	/23
255.255.255.0	/24
255.255.255.128	/25
255.255.255.192	/26
255.255.255.224	/27
255.255.255.240	/28
255.255.255.248	/29
255.255.255.252	/30

Class A
255.255.0.0 (/16)
Melakukan subnet terhadap subnet mask 255.0.0.0, terdapat 22 bit untuk di subnetting
Harus meninggalakan 2 bit untuk pengalamatan host
Mask 255.255.0.0 dengan alamat class A menggunakan 8 bit subnet

Berapa Banyak Subnet ?

$$2^n-2 = 2^8-2 = 254$$

Berapa banyak host per subnet ?

$$2^n-2 = 2^{16}-2 = 65.534$$

Berapa jumlah subnet yang vailid ?

$$256 - 255 = 1$$

1;2;3; semua di oktet ke dua
Subnetnya akan menjadi 10.1.0.0 ; 10.2.0.0; 10.3.0.0 dan seterusnya sampai 10.254.0.0
Alamat broadcast untuk setiap subnet ?
Host host yang vailid ?

Alamat subnet 10.1.0 . 0 . . . . . .10.254. 0
Host Pertama 10.1.0 . 1 .. . . .10.254.0.1
Host terakhir 10.1.255.254 . . . . .10.254.255.254
Alamat Broadcast 10.1.255.255 . . .10.254.255.255

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Review

Class A
255.255.240.0 (/20)
Melakukan subnet terhadap subnet mask 255.255.240.0, terdapat 12 bit untuk di subnetting
Harus meninggalakan 2 bit untuk pengalamatan host

Berapa Banyak Subnet ?

$2^n-2 = 2^{12}-2 = 4094$

Berapa banyak host per subnet ?

$2^n-2 = 2^{12}-2 = 4094$

Berapa jumlah subnet yang vailid ?

$256 - 240 = 16$

Karena oktet kedua 255 atau semua bit subnet on
Kita dapat memulai di oketet ketiga dengan nol, selama ada i bit subnet yang on di subnet kedua

Maka subnet pertama yang vailid 10.0.16.0 ; 10..0.32.0; 10,0,48,0; dst 10,0,240,0

Namun 10,0,0,0 dan 10,255,240,0 secara defaul bukan alamat subnet yang vailid

Barisan subnet 10,1,0,0;10,1,16,0; 10,1,32,0; 10,1,48,0 dst sampai 10,1,240,0

Perhatiakan kita dapat menggunakan 240 dioktet ketiga selama tidak semua bit subnet di oktet kedua on (bernilai 1) atau Ada 1 bit subnet di oktet kedua yang off (bernialai 0).
Kita dapat menggunakan 0 di oktet ke tiga selama oktet kedua tidak 0. atau 10,255,240,0 adalah tidak vailid karena semua bit on
Subnet yang terakhir yang vailid adalah 10,255,244,0

Alamat broadcast untuk setiap subnet ?
Host host yang vailid ?

Alamat subnet	10.1.0 . 0	10.1.16.0	. . .10.255.224. 0
Host Pertama	10.1.0 . 1	10.1.16.1	. . .10.255.224. 1
Host terakhir	10.1.15.254	10.1.31.254	. . .10.255.239.254
Alamat Broadcast	10.1.15.255	10.1.32.255	. . .10.255.239.255

Class A
255.255.255,192 (/26)

Berapa Banyak Subnet ?

$2^n-2 = 2^{18}-2 = 262,142$

Berapa banyak host per subnet ?

$2^n-2 = 2^6-2 = 62$

Berapa jumlah subnet yang vailid ?

$256 - 192 = 64$

Alamat broadcast untuk setiap subnet ?
Host host yang vailid ?

Alamat subnet	10.0.0.64	10.0,0,128	10,0,0,192	10,1,0,0
Host Pertama	10.0.0.65	10.0,0,129	10,0,0,193	10,1,0,1
Host terakhir	10.0.0.126	10.0,0,190	10,0,0,254	10,1,0,62
Alamat Broadcast	10.0.0.127	10.0,0,191	10,0,0,255	10,1,0,63

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Variable Length Subnet Mask (VLSM)

- To increase usability of Subnets
- Using classical subnetting the subnets are of same size
- Using VLSM the subnets can be of VARIABLE SIZE

Ex :

A Class C, NW need to be divided into 110,45,50 Hosts?

Available Options 1 :

- Natural NW Mask- Class C is 255.255.255.0 Which will provide nothin but 254 hosts under 1 NW

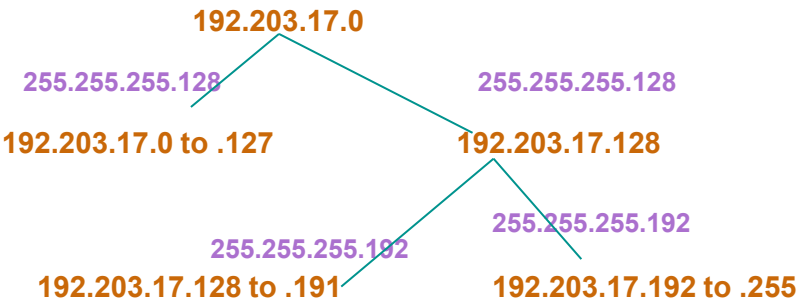
A Class C, NW need to be divided into 110,45,50 Hosts?

Available Options 2 :- The Subnet Mask of the form 255.255.255.X

X (in Binary)	X	No of Subnet	No of Host
1000 0000	128	2	128
1100 0000	192	4	64
1110 0000	224	8	32
1111 0000	240	16	16
1111 1000	248	32	8
1111 1100	252	64	4

A Class C, NW need to be divided into 110,45,50 Hosts?

> Hierarchical Subnetting



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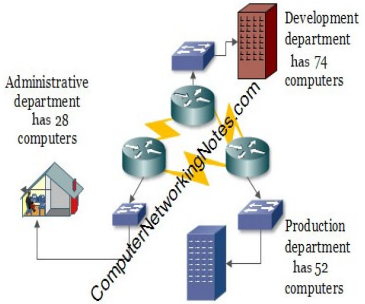
Ex2:

For example, assume that you are a network administrator at Laxmisoftwares. Company has three departments connected with wan links.

- Development department has 74 computers.
- Production department has 52 computers.
- Administrative department has 28 computers.
- All departments are connected with each other via wan link.
- Each wan link requires two IP addresses.



With FLSM, to accumulate this requirement we have two choices, either purchase a class B IP address or purchase at least two class C IP addresses.

First choice (*purchase a class B IP address*)

172.168.1.0/23

Subnetting of this address would give us 128 subnets and 510 hosts in each subnet. Our network requires only 6 subnets and 160 addresses. We would have to pay for 65356 addresses while you need only 160 addresses. Every IP address adds more dollars in company bill. Would you consider this address space for company?

Second choice (*purchase at least two Class C IP addresses*)

192.168.1.0/25

192.168.2.0/26

Subnetting of first address 192.168.1.0/25 would give us 2 subnets and 126 hosts in each subnet.

Subnetting of second address 192.168.2.0/26 would give us 4 subnets and 62 hosts in each subnet.

Collectively we are getting 6 subnets and 500 hosts from these two address spaces. We are still wasting more than 300 IP address, and we would have to purchase two address spaces.

Variable Length Subnet Mask

Variable Length Subnet Mask (VLSM) extends classic Subnetting. VLSM is a process of breaking down subnets into the smaller subnets, according to the need of individual networks. In above example company has requirement of 6 subnets and 160 host addresses. With VSLM you can fulfill this requirement with single class C address space.

VLSM Subnetting

In VLSM Subnetting, we do Subnetting of subnets according the network requirement.

Steps for VLSM Subnetting

Find the largest segment. Segment which need largest number of hosts address.

- Do Subnetting to fulfill the requirement of largest segment.
- Assign the appropriate subnet mask for the largest segment.
- For second largest segments, take one of these newly created subnets and apply a different, more appropriate, subnet mask to it.
- Assign the appropriate subnet mask for the second largest segment.
- Repeat this process until the last network.

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Example VLSM

Now you know the steps of VLSM Subnetting. Let's understand it with above example. Our company requires 6 subnets and 160 hosts.

Step 1 :- Oder all segments according the hosts requirement (Largest to smallest).

Subnet	Segment	Hosts
1	Development	74
2	Production	52
3	Administrative	28
4	Wan link 1	2
5	Wan link 2	2
6	Wan link 3	2

Step 2 :- Do subnetting for largest segment. Our largest segment needs 74 host addresses. /25 provide us two subnets with 126 hosts in each subnet.

192.168.1.0/25

Subnet	Subnet 1	Subnet 2
Network ID	192.168.1.0	192.168.1.128
First host address	192.168.1.1	192.168.1.129
Last host address	192.168.1.126	192.168.1.254
Broadcast ID	192.168.1.127	192.168.1.255

Step 3 :- Assign subnet mask to the largest segment. As you can see in above table, subnet 1 fulfill our largest segment requirement. Assign it to our segment.

Segment	Development
Requirement	74
CIDR	/25
Subnet mask	255.255.255.128
Network ID	192.168.1.0
First hosts	192.168.1.1
Last hosts	192.168.1.126
Broadcast ID	192.168.1.127

Step 4 :- Do subnetting for second largest segment from next available subnet. Next segment requires 52 host addresses. Subnetting of /25 has given us two subnets with 126 hosts in each, from that we have assigned first subnet to development segment. Second segment is available, we would do subnetting of this. /26 provide us 4 subnets with 62 hosts in each subnet.

192.168.1.0/26

Subnet	Subnet 1	Subnet 2	Subnet 3	Subnet 4
Network ID	0	64	128	192
First address	1	65	129	193
Last address	62	126	190	254
Broadcast ID	63	127	191	255

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We cannot use subnet 1 and subnet 2 (address from 0 to 127) as they are already assigned to development department. We can assign subnet 3 to our production department.

Segment	Production
Requirement	52
CIDR	/26
Subnet mask	255.255.255.192
Network ID	192.168.1.128
First hosts	192.168.1.129
Last hosts	192.168.1.190
Broadcast ID	192.168.1.191

Step 5 :- Our next segment requires 28 hosts. From above subnetting we have subnet 3 and subnet 4 available. Do subnetting for the requirement of 28 hosts.

Subnet	Sub 1	Sub 2	Sub 3	Sub 4	Sub 5	Sub 6	Sub 7	Sub 8
Net ID	0	32	64	96	128	160	192	224
First Host	1	33	65	95	129	161	193	225
LastHost	30	62	94	126	158	190	222	254
Broadcast ID	31	63	95	127	159	191	223	255

Subnets 1 to 6 [address from 0 to 191] are already occupied by previous segments. We can assign subnet 7 to this segment.

Segment	Administrative
Requirement	28
CIDR	/27
Subnet mask	255.255.255.224
Network ID	192.168.1.192
First hosts	192.168.1.193
Last hosts	192.168.1.222
Broadcast ID	192.168.1.223

Step 6 :- Our last three segments require 2 hosts per subnet. Do subnetting for these.

192.168.1.0/30
Valid subnets are:-
0,4,8,12,16,20,24,28,32,36,40,44,48,52,56,60,64,68,72,76,80,84,88,92,96,100,104,108,112,116,120,124,128,132,136,140,144,148,152,156,160,164,168,172,176,180,184,188,192,196,200,204,208,212,216,220,224,228,232,236,240,244,248,252,256

From these subnets, subnet 1 to subnet 56 (Address from 0 - 220) are already assigned to previous segments. We can use 224,228, and 232 for wan links.

Subnet	Subnet 57	Subnet 58	Subnet 59
Network ID	224	228	232
First host	225	229	233
Last host	226	230	234
Broadcast ID	227	231	235

Assign these subnets to wan links.
Wan Link 1

Segments	Wan Link 1
Requirement	2
CIDR	/30
Subnet mask	255.255.255.252
Network ID	192.168.1.224
First hosts	192.168.1.225
Last hosts	192.168.1.226
Broadcast ID	192.168.1.227

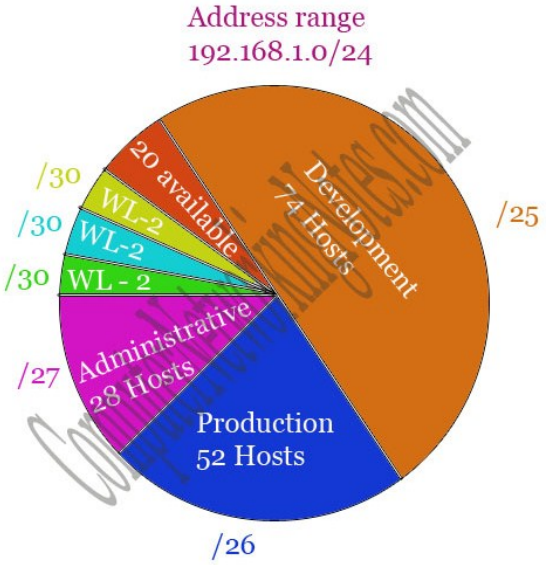
Wan Link 2

Segments	Wan Link 2
Requirement	2
CIDR	/30
Subnet mask	255.255.255.252
Network ID	192.168.1.228
First hosts	192.168.1.229
Last hosts	192.168.1.230
Broadcast ID	192.168.1.231

Wan Link 3

Segments	Wan Link 3
Requirement	2
CIDR	/30
Subnet mask	255.255.255.252
Network ID	192.168.1.232
First hosts	192.168.1.233
Last hosts	192.168.1.234
Broadcast ID	192.168.1.235

We have assigned IP addresses to all segments, still we have 20 addresses available. This is the magic of VLSM.



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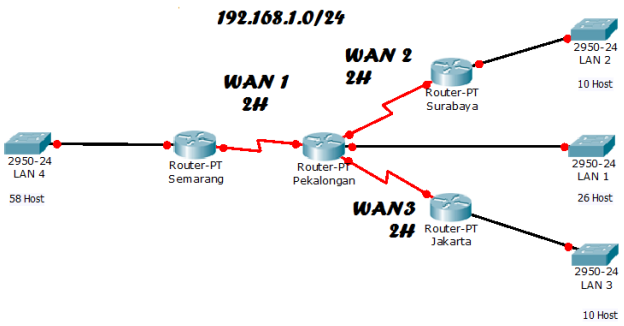
SUBNETTING DENGAN VLSM
Variable Length Subnet Mask

VLSM (Variable Length Subnet Mask) adalah sebuah cara pengelolaan pengalamatan IP yang lebih terstruktur dibandingkan sekedar menggunakan ***FLSM (Fixed Length Subnet Mask)***.

Perhitungan IP Address menggunakan metode VLSM adalah metode yang berbeda dengan memberikan suatu Network Address lebih dari satu subnet mask, berbeda jika menggunakan CIDR dimana suatu Network ID hanya memiliki satu subnet mask saja.

VLSM memiliki manfaat untuk mengurangi jumlah alamat yang terbuang. Untuk lebih mudahnya sebagai contoh, kita akan menghitung alamat IP menggunakan VLSM dengan topologi sebagai berikut:

CIDR = Classes Inter Domain Routing



Pertama, kita cari host yang paling banyak digunakan. yaitu pada LAN4 dengan 58 Host, LAN1 (26 Host), LAN2 (10 Host), LAN3 (10 Host), dan masing2 WAN 2 Host. Disini diberikan IP 192.168.1.0/24, dan kita akan membaginya dengan VLSM.

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NetMask Desimal	Netmask Biner	Format CIDR	Jumlah Host
255.255.255.0	**11111111.11111111.11111111.00000000**	/24	254
255.255.255.128	**11111111.11111111.11111111.10000000**	/25	126
255.255.255.192	**11111111.11111111.11111111.11000000**	/26	62
255.255.255.224	**11111111.11111111.11111111.11100000**	/27	30
255.255.255.240	**11111111.11111111.11111111.11110000**	/28	14
255.255.255.248	**11111111.11111111.11111111.11111000**	/29	6
255.255.255.252	**11111111.11111111.11111111.11111100**	/30	2

1. Menghitung IP untuk LAN4 (58 Host)

NetMask Desimal	Netmask Biner	Format CIDR	Jumlah Host
255.255.255.0	**11111111.11111111.11111111.00000000**	/24	254
255.255.255.128	**11111111.11111111.11111111.10000000**	/25	126
255.255.255.192	**11111111.11111111.11111111.11000000**	/26	62
255.255.255.224	**11111111.11111111.11111111.11100000**	/27	30
255.255.255.240	**11111111.11111111.11111111.11110000**	/28	14
255.255.255.248	**11111111.11111111.11111111.11111000**	/29	6
255.255.255.252	**11111111.11111111.11111111.11111100**	/30	2

Jika kita menggunakan /24 tentunya terlalu banyak Host yang tersisa (tdk digunakan),
Karena kita hanya butuh 58 Host. Kita tentukan subnet mask yang memiliki host lebih dari 58, dilihat dari tabel diatas yang terpenuhi adalah /26 (62 Host) dengan subnet 255.255.255.192

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berikut adalah peluang alamat IP yang digunakan dari /26
(Perhatikan 2-bit MSB pada oktet terakhir)

/26 = **11111111.11111111.11111111.11000000**

Network	IP Range	Broadcast
.0	1-62	.63
.64	65 - 126	.127
.128	129 - 190	.191
.192	193 - 254	.255

Untuk 58 Host kita menggunakan IP Address 192.168.1.0/26
Network 192.168.1.0
IP Range 192.168.1.1-192.168.1.62
Broadcast 192.168.1.63

2. Menghitung IP untuk LAN1 (26 Host)

Kita tentukan subnet mask yang memiliki 26 host lebih, dilihat dari tabel subnetting diatas yang terpenuhi adalah /27 (30 Host) dengan subnet 255.255.255.224.
Karena diLAN4 telah menggunakan IP 192.168.1.0/26 , maka kita akan menggunakan IP dibawahnya yang belum digunakan yaitu 192.168.1.64/26. seperti cara sebelumnya kita akan mengubah subnet mask nya menjadi 255.255.255.224.

NetMask Desimal	Netmask Biner	Format CIDR	Jumlah Host
255.255.255.0	**11111111.11111111.11111111.00000000**	/24	254
255.255.255.128	**11111111.11111111.11111111.10000000**	/25	126
255.255.255.192	**11111111.11111111.11111111.11000000**	/26	62
255.255.255.224	**11111111.11111111.11111111.11100000**	/27	30
255.255.255.240	**11111111.11111111.11111111.11110000**	/28	14
255.255.255.248	**11111111.11111111.11111111.11111000**	/29	6
255.255.255.252	**11111111.11111111.11111111.11111100**	/30	2

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berikut kemungkinan IP yang digunakan (/27):
11111111.11111111.11111111.11100000

Network	IP Range	Broadcast
.64	65-94	.95
.96	97-126	.127
.128	129-158	.159
.160	161-190	.191

Untuk 26 Host kita menggunakan IP Address 192.168.1.0/27
Network 192.168.1.64
IP Range 192.168.1.65-192.168.1.94
Broadcast 192.168.1.95

3. Menghitung IP untuk LAN3 (10 Host)

NetMask Desimal	Netmask Biner	Format CIDR	Jumlah Host
255.255.255.0	**11111111.11111111.11111111.00000000**	/24	254
255.255.255.128	**11111111.11111111.11111111.10000000**	/25	126
255.255.255.192	**11111111.11111111.11111111.11000000**	/26	62
255.255.255.224	**11111111.11111111.11111111.11100000**	/27	30
255.255.255.240	**11111111.11111111.11111111.11110000**	/28	14
255.255.255.248	**11111111.11111111.11111111.11111000**	/29	6
255.255.255.252	**11111111.11111111.11111111.11111100**	/30	2

Kita tentukan subnet mask yang memiliki 10 host lebih, dilihat dari tabel subnetting diatas yang terpenuhi adalah /28 (14 Host) dengan subnet 255.255.255.240.
Karena diLAN1 telah menggunakan IP 192.168.1.64/27 , maka kita akan menggunakan IP dibawahnya yang belum digunakan yaitu 192.168.1.96/27. seperti cara sebelumnya kita akan merubah subnet mask nya menjadi 255.255.255.240.

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berikut kemungkinan IP yang digunakan (/28):

11111111.11111111.11111111.11110000

Network	IP Range	Broadcast
.96	.97-.110	.111
.112	.113-.126	.127
.128	.129-.142	.143
.144	.145-.158	.159

Karena ada 2 LAN yang butuh 10 Host kita menggunakan IP address 192.168.1.96/28 dan 192.168.1.112/28
Network 192.168.1.96
IP Range 192.168.1.97-192.168.1.110
Broadcast 192.168.1.111

Network 192.168.1.112
IP Range 192.168.1.113-192.168.1.126
Broadcast 192.168.1.127

4. Menghitung WAN untuk LAN2 dan LAN3 (2 Host)

Kita tentukan subnet mask yang memiliki 2 host atau lebih, dilihat dari tabel subnetting diatas yang terpenuhi adalah /30 (2 Host) dengan subnet 255.255.255.252. Karena diLAN sebelumnya telah menggunakan IP 192.168.1.96/28 dan 192.168.1.112/28, maka kita akan menggunakan IP dibawahnya yang belum digunakan yaitu 192.168.1.128/28. seperti cara sebelumnya kita akan merubah subnet mask nya menjadi 255.255.255.252.

berikut kemungkinan IP yang digunakan (/30):

11111111.11111111.11111111.11111100

Network	IP Range	Broadcast
.128	.129-.130	.131
.132	.133-.134	.135
.136	.137-.138	.139
.140	.141-.142	.143
.144	.145-.146	.147

Karena ada 3 WAN yang butuh 2 Host kita menggunakan IP address 192.168.1.128/30, 192.168.1.132/30 dan 192.168.136/30
Network 192.168.1.128
IP Range 192.168.1.129-192.168.1.130
Broadcast 192.168.1.131

Network 192.168.1.132
IP Range 192.168.1.133-192.168.1.134
Broadcast 192.168.1.135

Network 192.168.1.136
IP Range 192.168.1.137-192.168.1.138
Broadcast 192.168.1.139

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Variable Length Subnet Mask (VLSM)

- Variable-Length Subnet Masking (VLSM) is the more realistic way of subnetting a network to make for the most efficient use of all of the bits.
- Classful (classical) subnetting are inefficient because all subnets have the same number of hosts because they all use the same subnet mask.
- For example, if you borrow 4 bits on a Class C network, you end up with 14 valid subnets of 14 valid hosts.

- A serial link to another router only needs 2 hosts, but with classical subnetting, you end up wasting 12 of those hosts.
- VLSM is the process of "subnetting a subnet" and using different subnet masks for different networks in your IP plan.

IP Subnet Zero

- With classical subnetting, you always have to eliminate the subnets that contain either all zeros or all ones in the subnet portion.
- Number of valid subnets $=2^N - 2$
- Cisco devices can use those subnets, as long as the command (ip subnet-zero) is in the configuration. **(Router(config)#ip subnet-zero).**

Hence, number of valid subnets $=2^N$

2^N	Number of total subnets created	
~~$2^N - 2$~~	~~Number of valid subnets created~~	No longer needed because you have the **ip subnet-zero** command enabled
2^H	Number of total hosts per subnet	
$2^H - 2$	Number of valid hosts per subnet	

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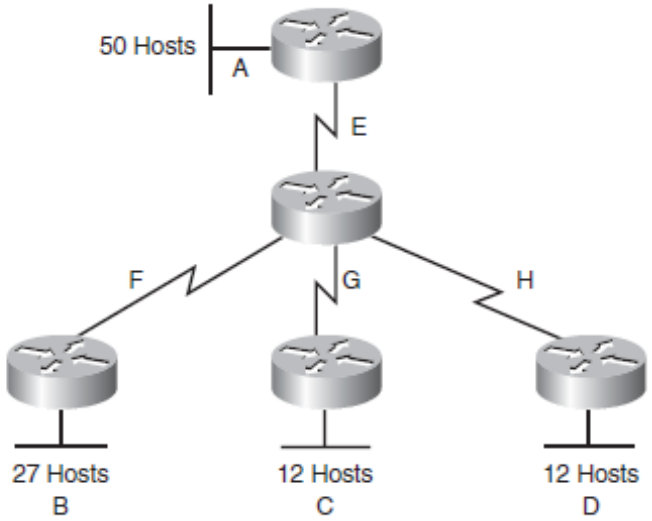
Example: Create an IP plan using VLSM for a Class C network—192.168.100.0/24

1) Determine how many H bits will be needed to satisfy the *largest network.*
2) Pick a subnet for the largest network to use.
3) Pick the next largest network to work with.
4) Pick the third largest network to work with.
5) Determine network numbers for serial links.



1) Determine how many H bits will be needed to satisfy the *largest network.*

- $2^H - 2 \geq 50$, Therefore H = 6 for Network A
- You need 6 H bits, 2 N bits (8 – 6) to create subnets

  Now have: NNHHHHHH (The 8 bits in the fourth octet)
- All subnetting will now have to start at this reference point.

2) Pick a subnet for the largest network to use.

- We have 2^N or 2^2 or 4 subnets:

NN = 00HHHHHH
01HHHHHH
10HHHHHH
11HHHHHH

- If you add all zeros to the H bits, you are left with the network numbers for the four subnets:

00**000000 = .0**
01**000000 = .64**
10**000000 = .128**
11**000000 = .192**

Network Address for all subnets

Network A

- **All of these subnets will have the same subnet mask:**

  11111111.11111111.11111111.11000000
- **Or 255.255.255.192 or /26 , The /x notation represents how to show different subnet masks when using VLSM.**
- **/26 means that the first 26 bits of the address are network; the remaining 6 bits are H bits.**

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3) Pick the next largest network to work with.

$2^H - 2 \geq 27$, H = 5, You need 5 H bits for Network B.

- Pick one of the remaining /26 networks to work with Network B.

  Select 128/26 network: 10**000000**

- But you need only 5 H bits, not 6. Therefore, you are left with 10**N00000**
- Where 10 represents the original pattern of subnetting.
- **N represents the extra bit.**
- **00000 represents the 5 H bits you need for Network B.**
- Because you have extra bit, you can create two smaller subnets from the original subnet: 10**000000 =.128 and** 10**100000 =.160**

3) Pick the next largest network to work with.

$2^H - 2 \geq 27$, H = 5, You need 5 H bits for Network B.

- **You have now subnetted a subnet! This is the basis of VLSM.**
- The mask now equals:

  11111111.11111111.11111111.11100000 or 255.255.255.224 or /27

- Pick one of these new sub-subnets for Network B: 10**000000 /27 = Network B**
- Use the remaining sub-subnet for future growth,
- You want to make sure the addresses are not overlapping with each other. So go back to the original table.

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3) Pick the next largest network to work with. (Continue)

00000000 =	.0/26	
01000000 =	.64/26	Network A
10000000 =	.128/26	
11000000 =	.192/26	

You can now break the .128/26 network into two smaller /27 networks and assign Network B.

00000000 =	.0/26	
01000000 =	.64/26	Network A
10000000 =	.128/26	Cannot use because it has been subnetted
10000000 =	.128/27	Network B
10100000 =	.160/27	
11000000 =	.192/26	

The remaining networks are still available to be assigned to networks or subnetted further for better efficiency.

4) Pick the third largest network to work with.

- $2^H - 2 \geq 12$, H = 4, Networks C and Network D = 12 hosts each.
- You can chose .192/26 network, or .160/27 network. (select 160/27): 101**00000**
- But you only need 4 H bits, not 5. Therefore, you are left with 101**N0000**
- Because you have this extra bit, you can create two smaller subnets from the original subnet: 101**00000 = .160 and** 101**10000 = .176,** Masks will be
- 11111111.11111111.11111111.11110000 or 255.255.255.240 or /28
- Pick one of these new sub-subnets for Network C and one for Network D.

4) Pick the third largest network to work with.

00000000 =	.0/26	
01000000 =	.64/26	Network A
10000000 =	.128/26	Cannot use because it has been subnetted
10000000 =	.128/27	Network B
10100000 =	.160/27	Cannot use because it has been subnetted
10100000	.160/28	Network C
10110000	.176/28	Network D
11000000 =	.192/26	

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5) Determine network numbers for serial links.

- All serial links between routers have the same property in that they only need two addresses in a network—one for each router interface.
- Determine the number of H bits needed for these networks: $2^H - 2 \geq 2$, H = 2
- You need 2 H bits to satisfy the requirements of Networks E, F, G, and H.
- You have two of the original subnets left to work with.

  Select the .0/26 network: 00**000000**

5) Determine network numbers for serial links.

- But you need only 2 H bits, not 6. Therefore, you are left with 00**NNNN00** where

  00 represents the original pattern of subnetting.

  NNNN represents the extra bits you have.

  00 represents the 2 H bits you need for the serial links.

- Because you have 4 **N bits, you can create 16 sub-subnets from the original subnet:**

  00**000000** = **.0/30**

  00**000100** = **.4/30**

  00**001000** = **.8/30**

  00**001100** = **.12/30**

  00**010000** = **.16/30**

  .....

  .....

  00**111000** = **.56/30**

  00**111100** = **.60/30**

- **You need only four of them. You can hold the rest for future expansion**
- **Going back to the original table**

Important Notice

- **No subnet number is used twice.**
- **You have now created an IP plan for the network and have made the plan as efficient as possible,**
- **wasting no addresses in the serial links and leaving room for future growth.**
- **This is the power of VLSM!**

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00000000 =	.0/26	Cannot use because it has been subnetted
00000000 =	.0/30	Network E
00000100 =	.4/30	Network F
00001000 =	.8/30	Network G
00001100 =	.12/30	Network H
00010000 =	.16/28	Future growth
01000000 =	.64/26	Network A
10000000 =	.128/26	Cannot use because it has been subnetted
10000000 =	.128/27	Network B
10100000 =	160/27	Cannot use because it has been subnetted
10100000	160/28	Network C
10110000	176/28	Network D
11000000 =	.192/26	Future growth

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Kosong

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Review

IP Address Classes

Class A	1 – 127	(Network 127 is reserved for loopback and internal testing)		
		Leading bit pattern	0	00000000.00000000.00000000.00000000 Network . Host . Host . Host
Class B	128 – 191	Leading bit pattern	10	10000000.00000000.00000000.00000000 Network . Network . Host . Host
Class C	192 – 223	Leading bit pattern	110	11000000.00000000.00000000.00000000 Network . Network . Network . Host
Class D	224 – 239	(Reserved for multicast)		
Class E	240 – 255	(Reserved for experimental, used for research)		

Private Address Space

Class A	10.0.0.0 to 10.255.255.255
Class B	172.16.0.0 to 172.31.255.255
Class C	192.168.0.0 to 192.168.255.255

Default Subnet Masks

Class A	255.0.0.0
Class B	255.255.0.0
Class C	255.255.255.0

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Decimal To Binary Conversion

Use all 8 bits for each problem

128	64	32	16	8	4	2	1 = 255	
1	1	1	0	1	1	1	0	238
0	0	1	0	0	0	1	0	34
								123
								50
								255
								200
								10
								138
								1

Scratch Area

238
-128
110
-64
46
-32
14
-8
6
-4
2
-2
0

34
-32
2
-2
0

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ANDING With
Default subnet masks

Every IP address must be accompanied by a subnet mask. By now you should be able to look at an IP address and tell what class it is. Unfortunately your computer doesn't think that way. For your computer to determine the network and subnet portion of an IP address it must "AND" the IP address with the subnet mask.

Default Subnet Masks:

Class A	255.0.0.0
Class B	255.255.0.0
Class C	255.255.255.0

ANDING Equations:

1 AND 1 = 1
1 AND 0 = 0
0 AND 1 = 0
0 AND 0 = 0

Sample:

What you see...

IP Address: 192 . 100 . 10 . 33

What you can figure out in your head...

Address Class:	C
Network Portion:	**192 . 100 . 10** . 33
Host Portion:	192 . 100 . 10 . **33**

In order for you computer to get the same information it must AND the IP address with the subnet mask in binary.

	Network	Host	
IP Address:	11000000.01100100.00001010	00100001	(192 . 100 . 10 . 33)
Default Subnet Mask:	11111111.01111111.11111111	00000000	(255 . 255 . 255 . 0)
AND:	11000000.01100100.00001010	00000000	(192 . 100 . 10 . 0)

ANDING with the default subnet mask allows your computer to figure out the network portion of the address.

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ANDING With
Custom subnet masks

When you take a single network such as 192.100.10.0 and divide it into five smaller networks (192.100.10.16, 192.100.10.32, 192.100.10.48, 192.100.10.64, 192.100.10.80) the outside world still sees the network as 192.100.10.0, but the internal computers and routers see five smaller subnetworks. Each independent of the other. This can only be accomplished by using a custom subnet mask. A custom subnet mask borrows bits from the host portion of the address to create a subnetwork address between the network and host portions of an IP address. In this example each range has 14 usable addresses in it. The computer must still AND the IP address against the custom subnet mask to see what the network portion is and which subnetwork it belongs to.

IP Address: 192 . 100 . 10 . 0
Custom Subnet Mask: 255.255.255.240

Address Ranges:
192.10.10.0 to 192.100.10.15
192.100.10.16 to 192.100.10.31
192.100.10.32 to 192.100.10.47 (Range in the sample below)
192.100.10.48 to 192.100.10.63
192.100.10.64 to 192.100.10.79
192.100.10.80 to 192.100.10.95
192.100.10.96 to 192.100.10.111
192.100.10.112 to 192.100.10.127
192.100.10.128 to 192.100.10.143
192.100.10.144 to 192.100.10.159
192.100.10.160 to 192.100.10.175
192.100.10.176 to 192.100.10.191
192.100.10.192 to 192.100.10.207
192.100.10.208 to 192.100.10.223
192.100.10.224 to 192.100.10.239
192.100.10.240 to 192.100.10.255

	Network	Sub Network	Host	
IP Address:	11000000.01100100.00001010.	0010	0001	(192 . 100 . 10 . 33)
Custom Subnet Mask:	11111111.01111111.11111111.	1111	0000	(255 . 255 . 255 . 240)
AND:	11000000.01100100.00001010.	0010	0000	(192 . 100 . 10 . 32)

Four bits borrowed from the host portion of the address for the custom subnet mask.

The ANDING process of the four borrowed bits shows which range of IP addresses this particular address will fall into.

In the next set of problems you will determine the necessary information to determine the correct subnet mask for a variety of IP addresses.

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Problem 1

Number of needed subnets **14**
Number of needed usable hosts **14**
Network Address **192.10.10.0**

Address class C

Default subnet mask 255 . 255 . 255 . 0

Custom subnet mask 255 . 255 . 255 . 240

Total number of subnets 16

Total number of host addresses 16

Number of usable addresses 14

Number of bits borrowed 4

Show your work for Problem 1 in the space below.

	256	128	64	32	16	8	4	2	-	Number of Hosts
Number of Subnets -	2	4	8	16	32	64	128	256		
	128	64	32	16	8	4	2	1	-	Binary values
192 . 10 . 10 .	0	0	0	0	0	0	0	0		

Add the binary value numbers to the left of the line to create the custom subnet mask.

128
64
32
+16
240

16
-2
14

Observe the total number of hosts.

Subtract 2 for the number of usable hosts.

Problem 2

Number of needed subnets **1000**
Number of needed usable hosts **60**
Network Address **165.100.0.0**

Address class B

Default subnet mask 255 . 255 . 0 . 0

Custom subnet mask 255 . 255 . 255 . 192

Total number of subnets 1,024

Total number of host addresses 64

Number of usable addresses 62

Number of bits borrowed 10

Show your work for Problem 2 in the space below.

Number of Hosts -	65,536	32,768	16,384	8,192	4096	2048	1024	512	.	256	128	64	32	16	8	4	2	
Number of Subnets -	2	4	8	16	32	64	128	256.		512	1024	2048	4096	8,192	16,384	32,768	65,536	
Binary values -	128	64	32	16	8	4	2	1	.	128	64	32	16	8	4	2	1	
165 . 100 .	0	0	0	0	0	0	0	0	.	0	0	0	0	0	0	0	0	

Add the binary value numbers to the left of the line to create the custom subnet mask.

128
64
32
16
8
4
2
+1
255

128
+64
192

64
-2
62

Observe the total number of hosts.

Subtract 2 for the number of usable hosts.

13

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Problem 3

Network Address **148.75.0.0 /26**

/26 indicates the total number of bits used for the network and subnetwork portion of the address. All bits remaining belong to the host portion of the address.

Address class B

Default subnet mask 255 . 255 . 0 . 0

Custom subnet mask 255 . 255 . 255 . 192

Total number of subnets 1,024

Total number of host addresses 64

Number of usable addresses 62

Number of bits borrowed 10

Show your work for Problem 3 in the space below.

Number of Hosts –	65,536	32,768	16,384	8,192	4096	2048	1024	512	.	256	128	64	32	16	8	4	2	
Number of Subnets –									256.	512	1024	2048	4,096	8,192	16,384	32,768	65,536	
Binary values –	128	64	32	16	8	4	2	1	.	128	64	32	16	8	4	2	1	
148 . 75 .	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**.**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	

Add the binary value numbers to the left of the line to create the custom subnet mask.

```
  128   128
   64   +64
   32   ----
   16    192
    8
    4
    2
   +1
  ----
  255
```

```
  64
 -2
 ---
 62
```

Observe the total number of hosts.

Subtract 2 for the number of usable hosts.

```
 1024
   -2
 -----
 1,022
```

Subtract 2 for the total number of subnets to get the usable number of subnets.

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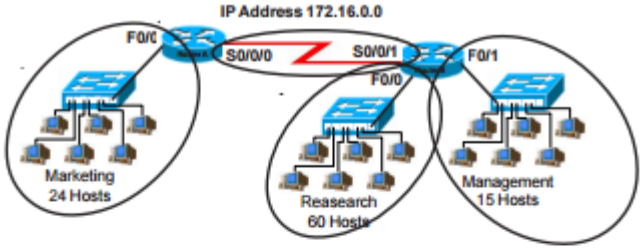
Subnetting 1 F0/1 F0/0 Router A S0/0/0 S0/0/1 Router B Based on the information in the graphic shown, design a network addressing scheme that will supply the minimum number of subnets, and allow enough extra subnets and hosts for 100% growth in both areas. Circle each subnet on the graphic and answer the questions below.



Address class	B
Custom subnet mask	255.255.224.0
Minimum number of subnets needed	4
Extra subnets required for 100% growth (Round up to the next whole number)	+ 4
Total number of subnets needed	= 8
Number of host addresses in the largest subnet group	60
Number of addresses needed for 100% growth in the largest subnet (Round up to the next whole number)	+ 60
Total number of address needed for the largest subnet	= 120

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

IP address range for Research	172.16.0.0 to 172.31.255
IP address range for Marketing	172.16.32.0 to 172.63.255
IP address range for Management	172.16.64.0 to 172.95.255
IP address range for Router A to Router B serial connection	172.16.96.0 to 172.127.255

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Number of Hosts –				65,536	32,768	16,384	8,192	4096	2048	1024	512	. 256	128	64	32	16	8	4	2
Number of Subnets –				2	4	8	16	32	64	128	256.	512	1024	2048	4096	8,192	16,384	32,768	65,536
Binary values –				128	64	32	16	8	4	2	1 .	128	64	32	16	8	4	2	1
172 . 16 .				**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0 .**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

(0)			0	172.16.0.0	to	172.16.31.255
(1)			1	172.16.32.0	to	172.16.63.255
(2)		1	0	172.16.64.0	to	172.16.95.255
(3)		1	1	172.16.96.0	to	172.16.127.255
(4)	1	0	0	172.16.128.0	to	172.16.159.255
(5)	1	0	1	172.16.160.0	to	172.16.191.255
(6)	1	1	0	172.16.192.0	to	172.16.223.255
(7)	1	1	1	172.16.224.0	to	172.16.255.255

$$\begin{array}{r} 4 \\ \times 10 \\ \hline 4 \end{array}$$

$$\begin{array}{r} 60 \\ \times 10 \\ \hline 60 \end{array}$$

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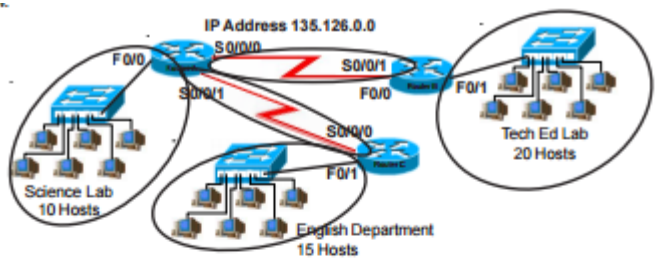
Subnetting 2 F0/0 S0/0/0 Router A S0/0/1 Based on the information in the graphic shown, design a network addressing scheme that will supply the minimum number of hosts per subnet, and allow enough extra subnets and hosts for 30% growth in all areas. Circle each subnet on the graphic and answer the questions below.



Address class	B
Custom subnet mask	255.255.255.224
Minimum number of subnets needed	5
Extra subnets required for 30% growth (Round up to the next whole number)	+ 2
Total number of subnets needed	= 7
Number of host addresses in the largest subnet group	20
Number of addresses needed for 30% growth in the largest subnet (Round up to the next whole number)	+ 6
Total number of address needed for the largest subnet	= 26

Start with the first subnet and arrange your sub-networks from the largest group to the smallest.

IP address range for Tech Ed	135.126.0.0 to 135.126.0.31
IP address range for English	135.126.0.32 to 135.126.0.63
IP address range for Science	135.126.0.64 to 135.126.0.95
IP address range for Router A to Router B serial connection	135.126.0.96 to 135.126.0.127
IP address range for Router A to Router B serial connection	135.126.0.128 to 135.126.0.159

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Number of Hosts -	65,536	32,768	16,384	8,192	4,096	2,048	1,024	512	. 256	128	64	32	16	8	4	2
Number of Subnets -	2	4	8	16	32	64	128	256.	512	1024	2048	4,096	8,192	16,384	32,768	65,536
Binary values -	128	64	32	16	8	4	2	1 .	128	64	32	16	8	4	2	1
135. 126 .	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0 .**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**

	(0)		.			0	135.126.0.0	to	135.126.0.31
	(1)					1	135.126.0.32	to	135.126.0.63
	(2)				1	0	135.126.0.64	to	135.126.0.95
5	(3)				1	1	135.126.0.96	to	135.126.0.127
x.3	(4)			1	0	0	135.126.0.128	to	135.126.0.159
15	(5)			1	0	1	135.126.0.160	to	135.126.0.191
(Round up to 2)	(6)			1	1	0	135.126.0.192	to	135.126.0.223
	(7)			1	1	1	135.126.0.224	to	135.126.0.255
	(8)	1	.	0	0	0	135.126.1.0	to	135.126.1.31
	(9)	1	.	0	0	1	135.126.1.32	to	135.126.1.63
20	(10)	1	.	0	1	0	135.126.1.64	to	135.126.1.95
x.3	(11)	1	.	0	1	1	135.126.1.96	to	135.126.1.127
6	(12)	1		1	0	0	135.126.1.128	to	135.126.1.159
	(13)	1	.	1	0	1	135.126.1.160	to	135.126.1.191
	(14)	1	.	1	1	0	135.126.1.192	to	135.126.1.223
	(15)	1	.	1	1	1	135.1261.224	to	135.126.1.255

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IP Address Breakdown

/24 8+8+8 255.255.255.0 256 Hosts	/25 8+8+8+1 255.255.255.128 128 Hosts	/26 8+8+8+2 255.255.255.192 64 Hosts	/27 8+8+8+3 255.255.255.224 32 Hosts	/28 8+8+8+4 255.255.255.240 16 Hosts	/29 8+8+8+5 255.255.255.248 8 Hosts	/30 8+8+8+6 255.255.255.252 4 Hosts
0-255	0-127	0-63		0-15	0-7	0-3
						4-7
					8-15	8-11
						12-15
				16-31	16-23	16-19
						20-23
					24-31	24-27
						28-31
				32-47	32-39	32-35
						36-39
					40-47	40-43
						44-47
				48-63	48-55	48-51
						52-55
					56-63	56-59
						60-63
		64-127		64-79	64-71	64-67
						68-71
					72-79	72-75
						76-79
				80-95	80-87	80-83
						84-87
					88-95	88-91
						92-95
				96-111	96-103	96-99
						100-103
					104-111	104-107
						108-111
				112-127	112-119	112-115
						116-119
					120-127	120-123
						124-127
	128-255	128-191		128-143	128-135	128-131
						132-135
					136-143	136-139
						140-143
				144-159	144-151	144-147
						148-151
					152-159	152-155
						156-159
				160-175	160-167	160-163
						164-167
					168-175	168-171
						172-175
				176-191	176-183	176-179
						180-183
					184-191	184-187
						188-191
		192-255		192-207	192-199	192-195
						196-199
					200-207	200-203
						204-207
				208-223	208-215	208-211
						212-215
					216-223	216-219
						220-223
				224-239	224-231	224-227
						228-231
					232-239	232-235
						236-239
				240-255	240-247	240-243
						244-247
					248-255	248-251
						252-255

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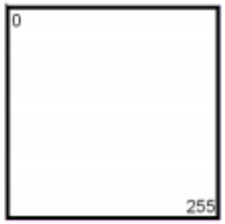
Visualizing Subnets Using The Box Method

The box method is the simplest way to visualize the breakdown of subnets and addresses into smaller sizes.

Start with a square. The whole square is a single subnet comprised of 256 addresses.

/24
255.255.255.0
256 Hosts
1 Subnet

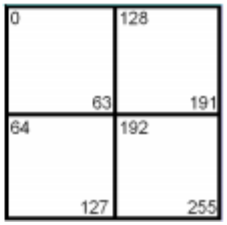


Split the box in half and you get two subnets with 128 addresses,

/25
255.255.255.128
128 Hosts
2 Subnets

Divide the box into quarters and you get four subnets with 64 addresses,

/26
255.255.255.192
64 Hosts
4 Subnets



Split each individual square and you get eight subnets with 32 addresses,

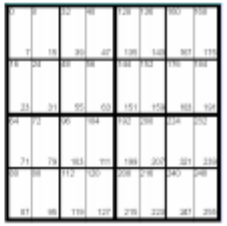
/27
255.255.255.224
32 Hosts
8 Subnets

Split the boxes in half again and you get sixteen subnets with sixteen addresses,

/28
255.255.255.240
16 Hosts
16 Subnets

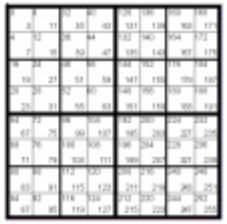
The next split gives you thirty two subnets with eight addresses,

/29
255.255.255.248
8 Hosts
32 Subnets



The last split gives sixty four subnets with four addresses each,

/30
255.255.255.252
4 Hosts
64 Subnets



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Class A Addressing Guide					
CIDR	# of Bits Borrowed	Subnet Mask	Total # of Subnets	Total # of Hosts	Usable # of Hosts
/8	0	255.0.0.0	1	16,777,216	16,777,214
/9	1	255.128.0.0	2	8,388,608	8,388,606
/10	2	255.192.0.0	4	4,194,304	4,194,302
/11	3	255.224.0.0	8	2,097,152	2,097,150
/12	4	255.240.0.0	16	1,048,576	1,048,574
/13	5	255.248.0.0	32	524,288	524,286
/14	6	255.252.0.0	64	262,144	262,142
/15	7	255.254.0.0	128	131,072	131,070
/16	8	255.255.0.0	256	65,536	65,534
/17	9	255.255.128.0	512	32,768	32,766
/18	10	255.255.192.0	1,024	16,384	16,382
/19	11	255.255.224.0	2,048	8,192	8,190
/20	12	255.255.240.0	4,096	4,096	4,094
/21	13	255.255.248.0	8,192	2,048	2,046
/22	14	255.255.252.0	16,384	1,024	1,022
/23	15	255.255.254.0	32,768	512	510
/24	16	255.255.255.0	65,536	256	254
/25	17	255.255.255.128	131,072	128	126
/26	18	255.255.255.192	262,144	64	62
/27	19	255.255.255.224	524,288	32	30
/28	20	255.255.255.240	1,048,576	16	14
/29	21	255.255.255.248	2,097,152	8	6
/30	22	255.255.255.252	4,194,304	4	2

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Class B Addressing Guide					
CIDR	# of Bits Borrowed	Subnet Mask	Total # of Subnets	Total # of Hosts	Usable # of Hosts
/16	0	255.255.0.0	1	65,536	65,534
/17	1	255.255.128.0	2	32,768	32,766
/18	2	255.255.192.0	4	16,384	16,382
/19	3	255.255.224.0	8	8,192	8,190
/20	4	255.255.240.0	16	4,096	4,094
/21	5	255.255.248.0	32	2,048	2,046
/22	6	255.255.252.0	64	1,024	1,022
/23	7	255.255.254.0	128	512	510
/24	8	255.255.255.0	256	256	254
/25	9	255.255.255.128	512	128	126
/26	10	255.255.255.192	1,024	64	62
/27	11	255.255.255.224	2,048	32	30
/28	12	255.255.255.240	4,096	16	14
/29	13	255.255.255.248	8,192	8	6
/30	14	255.255.255.252	16,384	4	2

Class C Addressing Guide					
CIDR	# of Bits Borrowed	Subnet Mask	Total # of Subnets	Total # of Hosts	Usable # of Hosts
/24	0	255.255.255.0	1	256	254
/25	1	255.255.255.128	2	128	126
/26	2	255.255.255.192	4	64	62
/27	3	255.255.255.224	8	32	30
/28	4	255.255.255.240	16	16	14
/29	5	255.255.255.248	32	8	6
/30	6	255.255.255.252	64	4	2

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IP Addresses

- Structure of an IP address
- Subnetting
- CIDR
- IP Version 6 addresses

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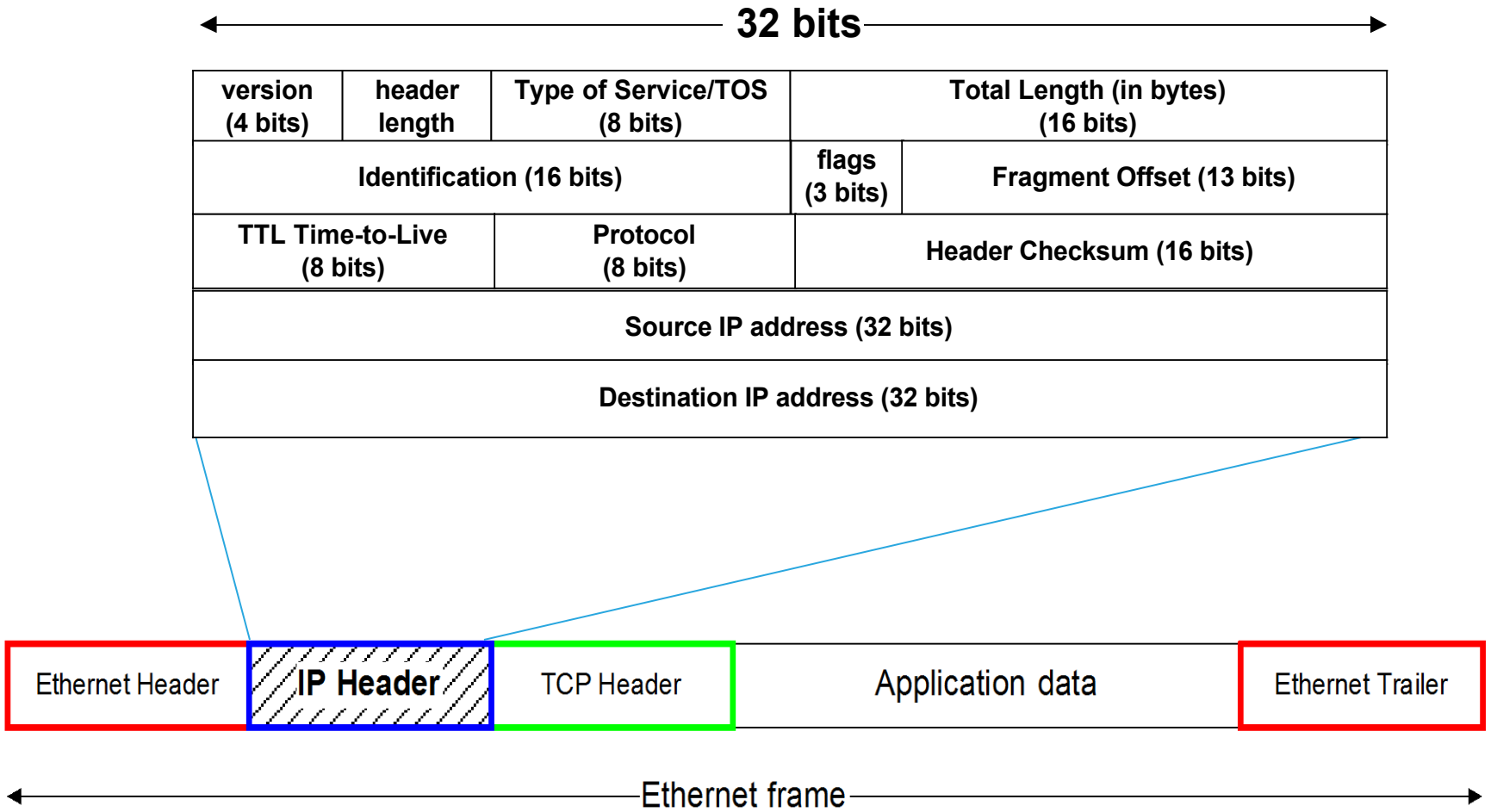
IP Addresses



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What is an IP Address?

- An IP address is a unique global address for a network interface

- An IP address:

  - is a **32 bit long** identifier

  - encodes a network number (**network prefix**)

    and a **host number**

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Dotted Decimal Notation

- IP addresses are written in a so-called ***dotted decimal* notation**
- Each byte is identified by a decimal number in the range [0..255]:

- **Example:**

10000000	10001111	10001001	10010000
1st Byte	2nd Byte	3rd Byte	4th Byte
= 128	**= 143**	**= 137**	**= 144**

128.143.137.144

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Network prefix and Host number

- The network prefix identifies a network and the host number identifies a specific host (actually, interface on the network).

- **How do we know how long the network prefix is?**

  The network prefix **used** to be implicitly defined (**class-based addressing, A,B,C,D…**)

  The network prefix now is flexible and is indicated by a **prefix/netmask (classless).**

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Example

Example: argon.cs.virginia.edu

- IP address is 128.143.137.144

  Is that enough info to route datagram??? -> No, need netmask or prefix at every IP device (host and router)

- Using Prefix notation IP address is: **128.143.137.144/16**

  Network prefix is 16 bits long

- Network mask is: **255.255.0.0** or hex format: **ffff0000**

  -----> **Network id** (IP address **AND** Netmask) is: **128.143.0.0**

  -----> **Host number** (IP address **AND** inverse of Netmask) is: **137.144**

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Subnetting

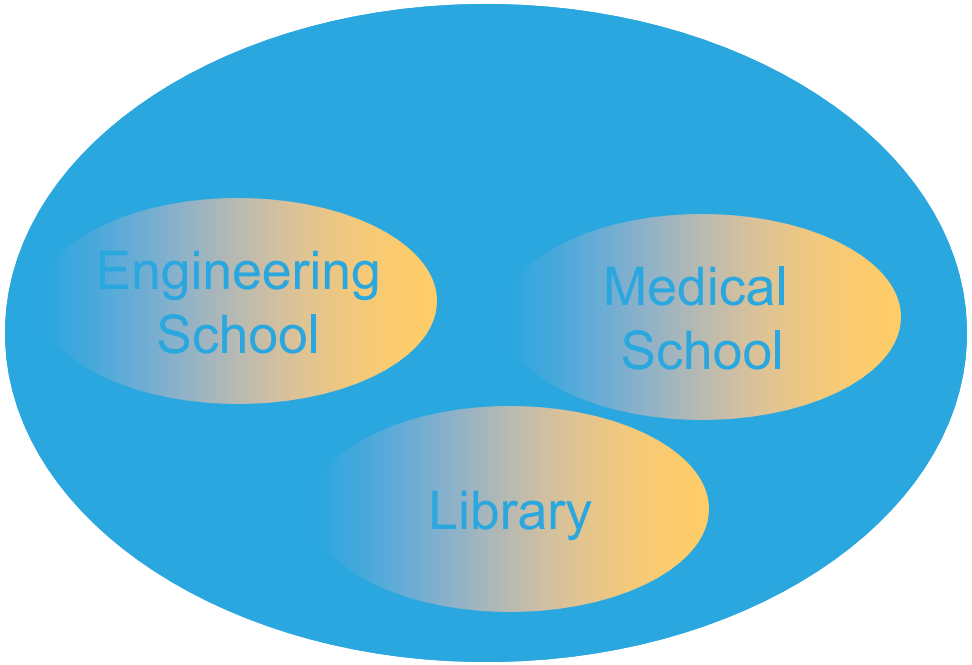
- **Problem**: Organizations have multiple networks which are independently managed

  Solution 1: Allocate an address for each network

  Difficult to manage

  From the outside of the organization, each network must be addressable ie have an identifiable address.

  Solution 2: Add another level of hierarchy to the IP addressing structure



Subnetting

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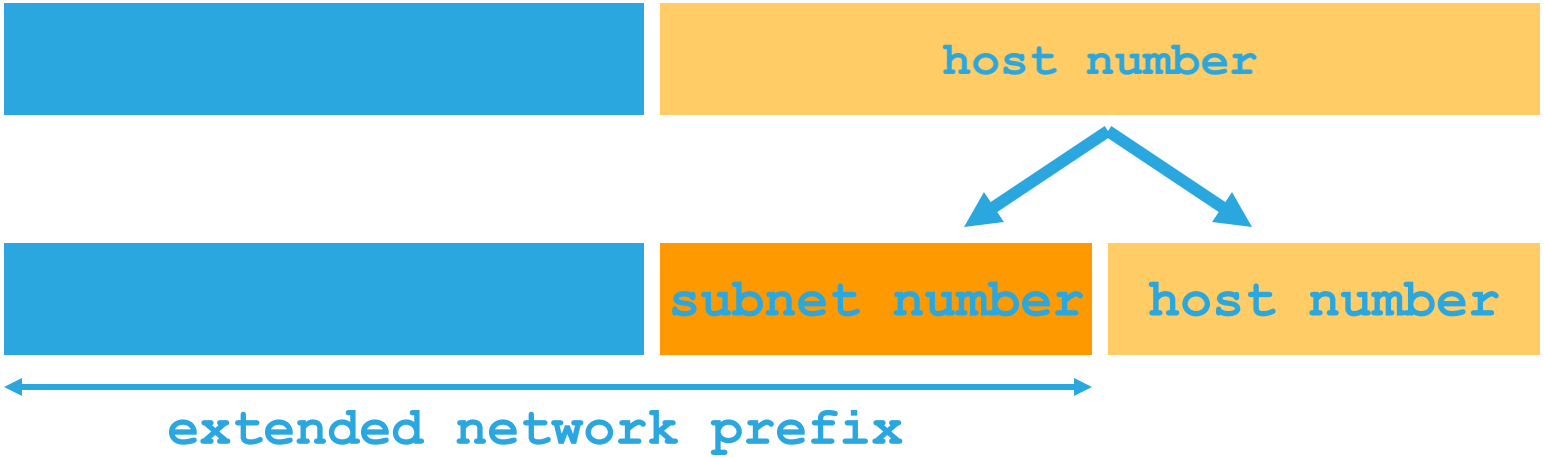
Basic Idea of Subnetting

- Split the host number portion of an IP address into a **subnet number** and a (smaller) **host number**.
- Result is a 3-layer hierarchy



- Then:
  - Subnets can be freely assigned within the organization
  - Internally, subnets are treated as separate networks
  - Subnet structure is not visible outside the organization

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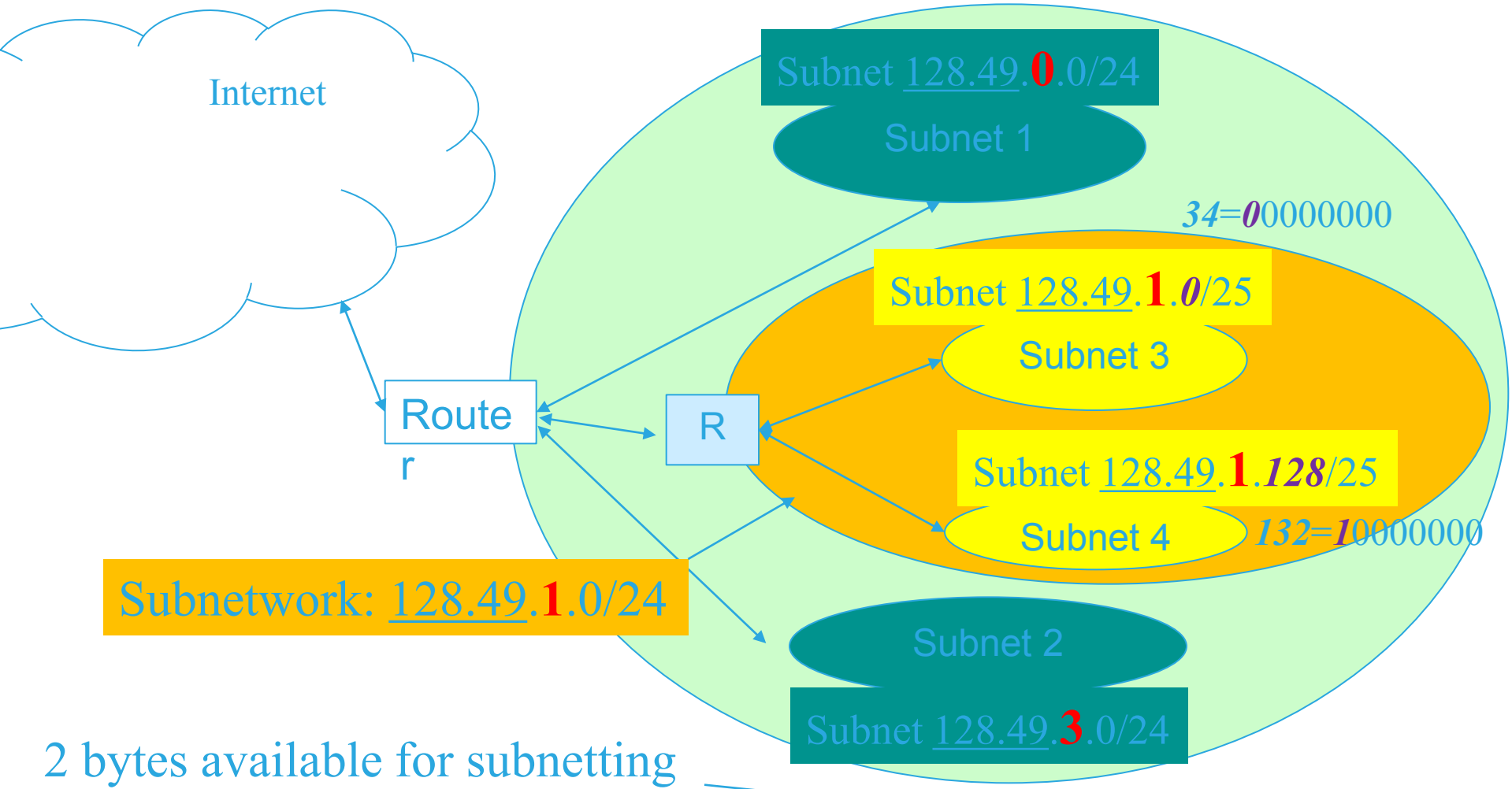
Example of a Subnetting Plan



2 bytes available for subnetting

IP Network: 128.49.0.0/16

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Advantages of Subnetting

- With subnetting, IP addresses use a 3-layer hierarchy:
  - Network
  - Subnet
  - Host
- Improves efficiency of IP addresses by not consuming an entire address space for each physical network.
- Reduces router complexity. Since external routers do not know about subnetting, the complexity of routing tables at external routers is reduced.
- Note: Length of the subnet mask need not be identical at all subnetworks.

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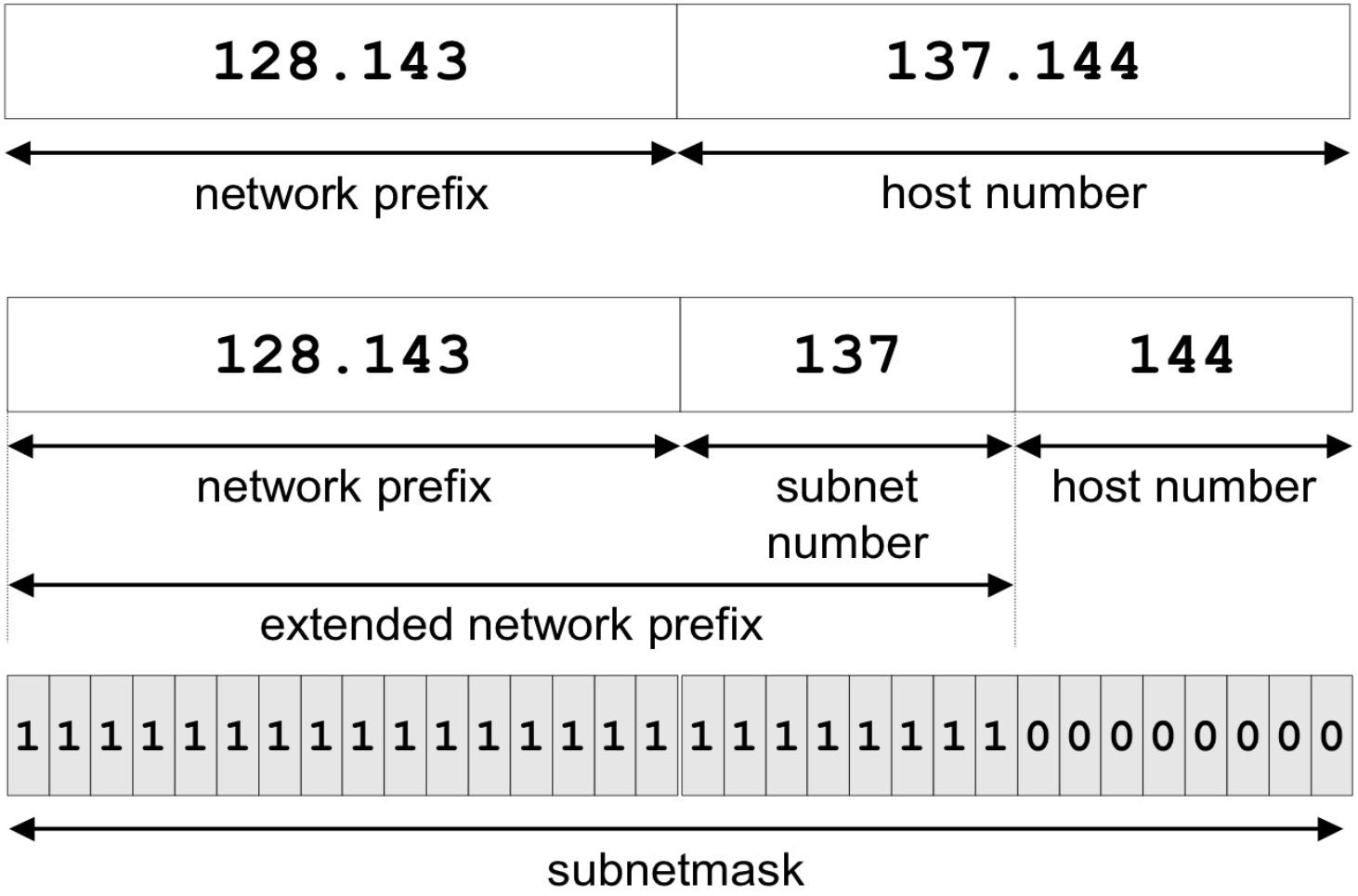
Subnetting Example: Argon



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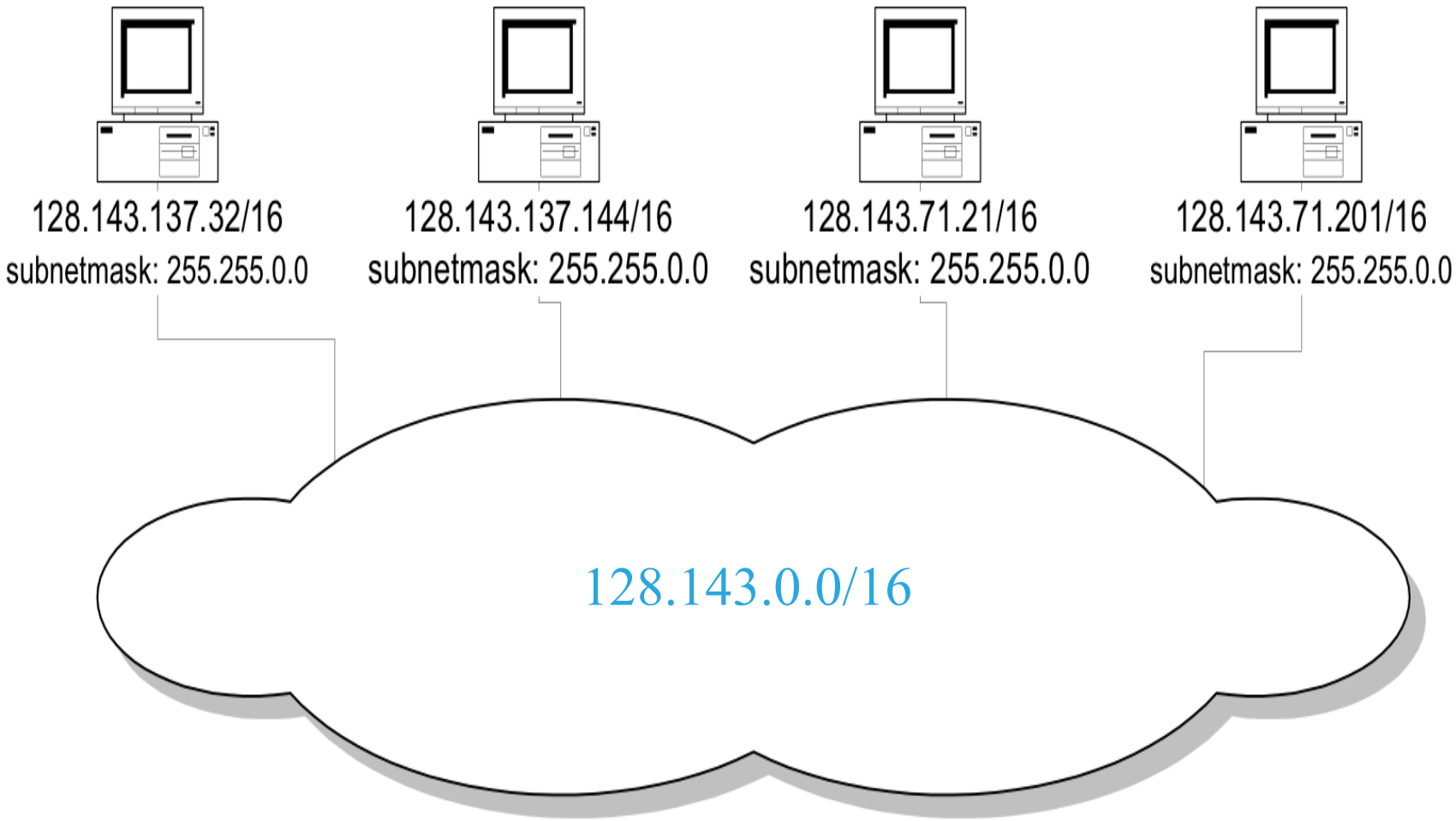
Network without subnets



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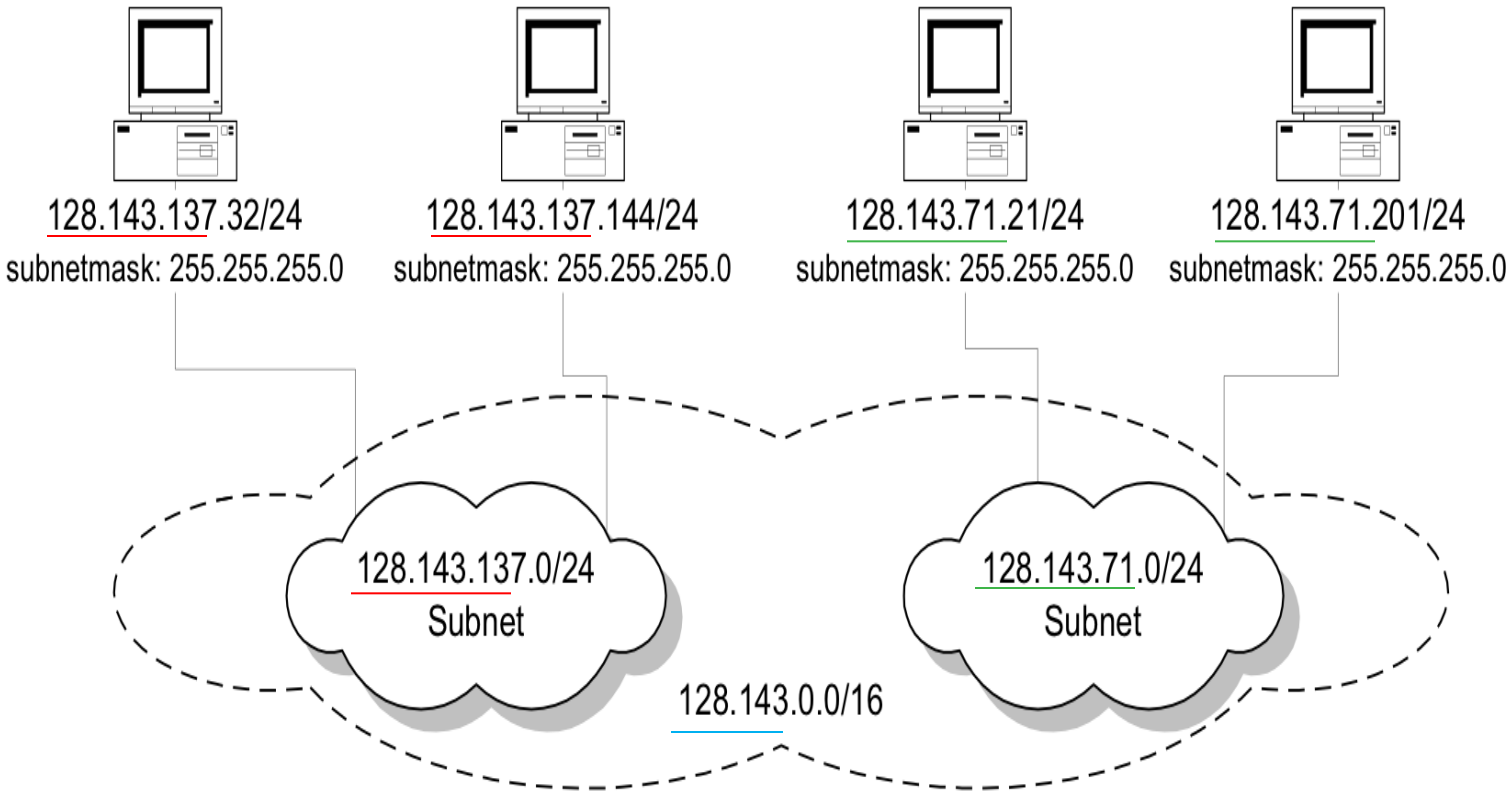
Same Network with Subnets



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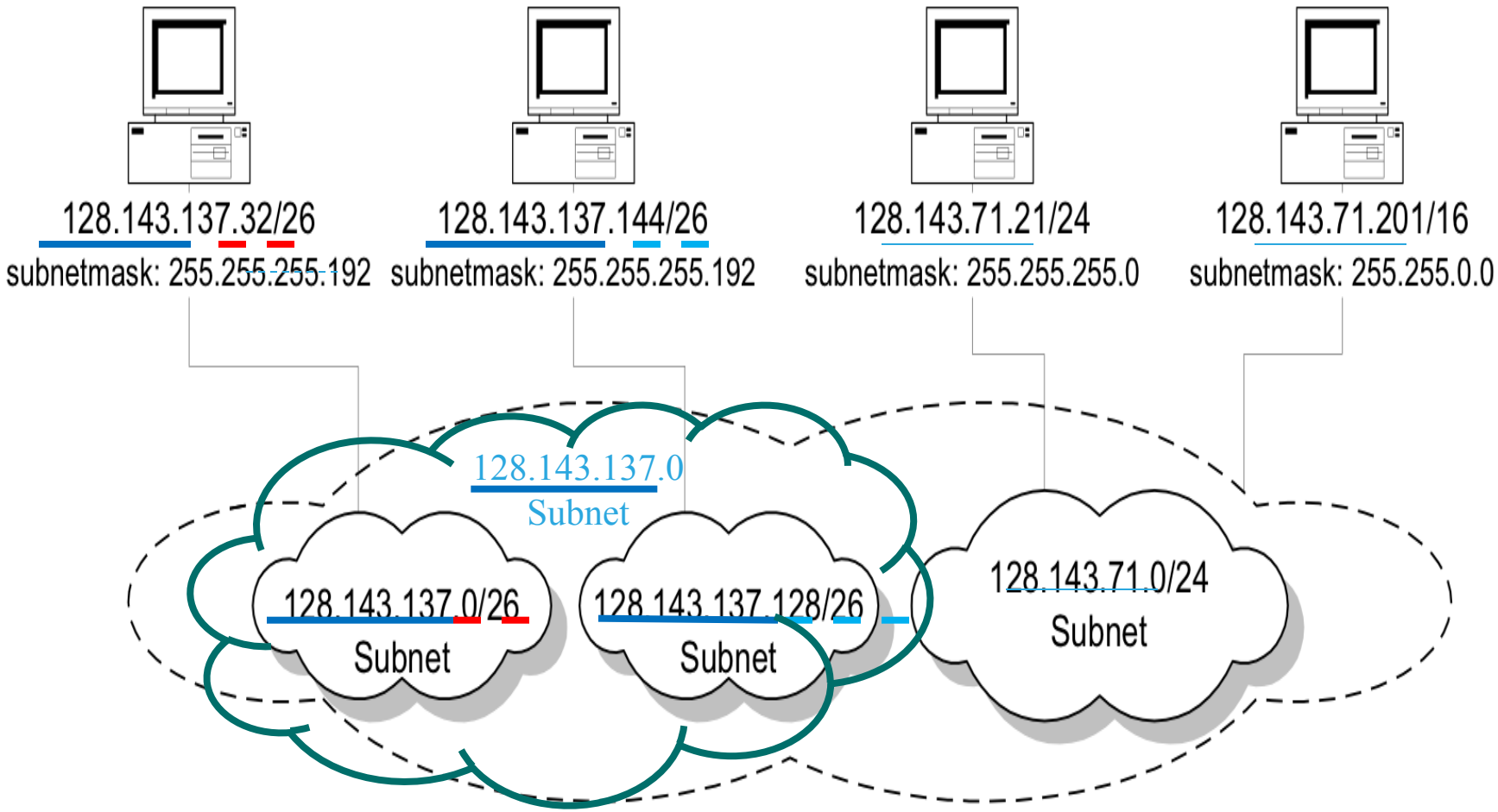
Same network with different subnetmasks



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Subnetting Example

- An organization with 4 departements has the following IP address space: 10.2.22.0/23. As the systems manager, you are required to create subnets to accommodate the IT needs of 4 departments. The subnets have to support to 200, 61, 55, and 41 hosts respectively. What are the 4 **subnet network** numbers?

- Solution:

  10.2.22.0/24 (256 addresses > 200)

  10.2.23.0/26 (64 addresses >61)

  10.2.23.64/26 (64 addresses > 55)

  10.2.23.128/26 (64 addresses > 41)

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CIDR - Classless Interdomain Routing

- **Goals:**

  Restructure IP address assignments to increase efficiency

  Hierarchical routing aggregation to minimize route table entries

Key Concept: The length of the network id (prefix) in IP addresses is arbitrary/flexible and is defined by the network hierarchy.

- Consequence:

  Routers use the IP address **and** the length of the prefix for forwarding.

  All advertised IP addresses must include a prefix

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CIDR Example

- CIDR notation of a network address:

  192.0.2.0/18

  "18" says that the first 18 bits are the network part of the address

- The network part is called the network prefix
- Example:

  Assume that a site requires an IP network domain that can support 1000 IP host addresses

  With CIDR, the network is assigned a continuous block of 1024 = 2^{10} (>1000) addresses with a 32-10 = 22-bit long prefix

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CIDR: Prefix Size vs. Host Space

CIDR Block Prefix	# of Host Addresses
/27	32 hosts
/26	64 hosts
/25	128 hosts
/24	256 hosts
/23	512 hosts
/22	1,024 hosts
/21	2,048 hosts
/20	4,096 hosts
/19	8,192 hosts
/18	16,384 hosts
/17	32,768 hosts
/16	65,536 hosts
/15	131,072 hosts

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CIDR and Address assignments

- Backbone ISPs obtain large blocks of IP address space and then reallocate portions of their address blocks to their customers.

Example:

- Assume that an ISP owns the address block 206.0.64.0/18, which represents 16,384 ($2^{32-18}=2^{14}$) IP host addresses
- Suppose a client requires 800 host addresses
  - $512=2^9<800<1024=2^{10}$ -> 32-10 = 22,
  - Assigning a /22 block, i.e., 206.0.68.0/22 -> gives a block of 1,024 (2^{10}) IP addresses to client.

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Subnetting and Classless Inter Domain Routing (CIDR)

- Subnetting is done by allocating some of the leading bits of the host number to indicate a subnet number.
  - With subnetting, the network prefix and the subnet number make up an extended network prefix.
  - The extended prefix can be expressed in terms of a subnetmask or, using CIDR notation, by adding the length of the extended subnetmask after the IP address.
- For example, for Argon, the first byte of the host number (the third byte of the IP address) is used to denote the subnet number.
  - 128.143.0.0/16 is the IP address of the network (network prefix /16),
  - 128.143.137.0/24 is the IP address of the subnet,
  - 128.143.137.144/32 is the IP address of the host, and
  - 255.255.255.0 is the subnetmask of the host (or subnet prefix /24))

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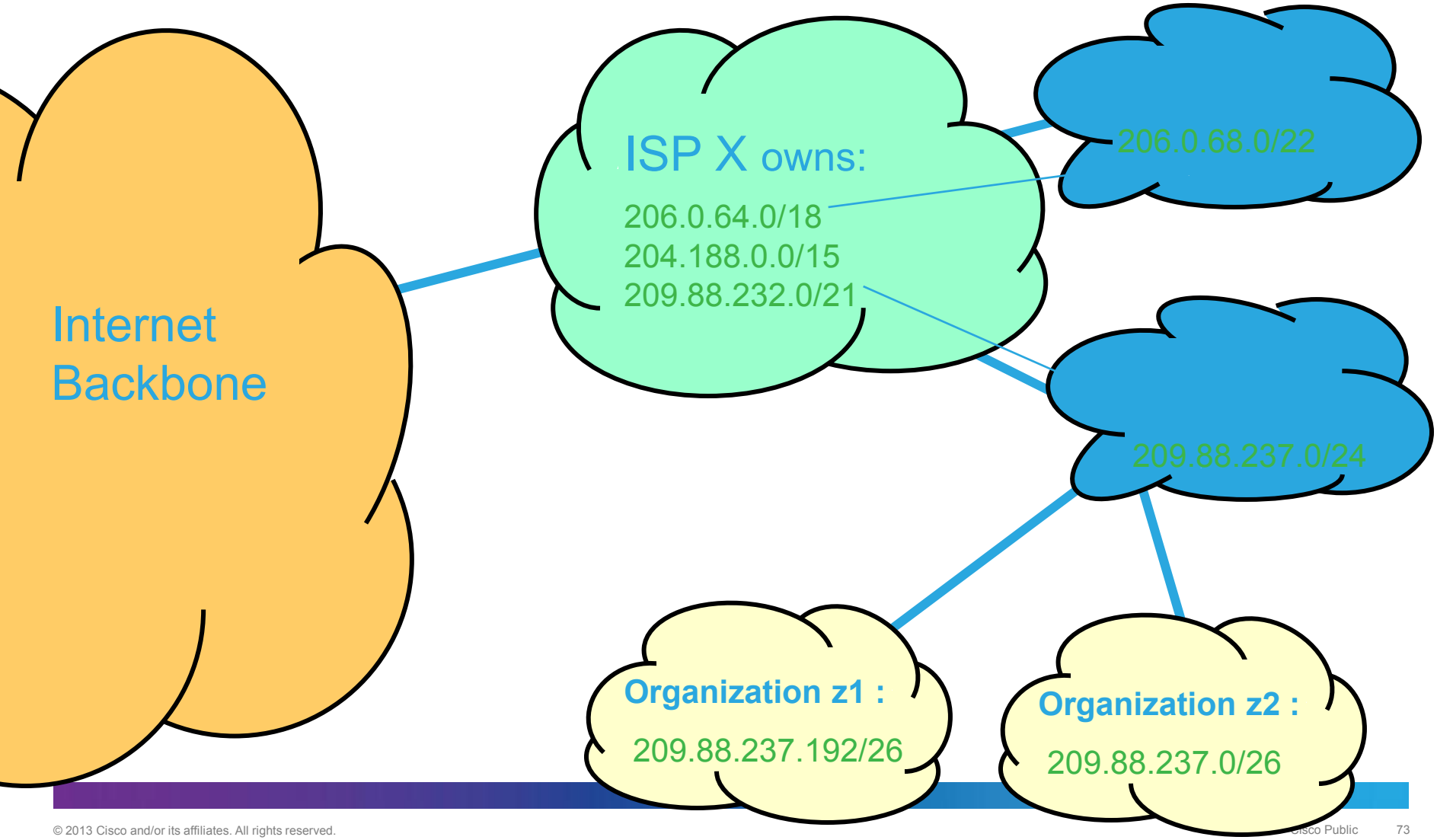
CIDR and Routing Information



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CIDR and Routing Information

Backbone routers do not know anything about Company X, ISP Y, or Organizations Z1, Z2.

ISP K does not know about Organizations Z1, Z2.

ISP K owns:

206.0.68.0/22

ISP Y sends everything which matches the prefix:
209.88.237.192/26 to Organizations Z1
209.88.237.0/26 to Organizations Z2

ISP K sends everything which matches the prefix:
206.0.68.0/22 to Company X,
209.88.237.0/24 to ISP Y

209.88.237.0/24

Backbone sends everything which matches the prefixes
206.0.64.0/18, 204.188.0.0/15,
209.88.232.0/21 to ISP K.

Organization Z1 :

209.88.237.192/26

Organization Z2 :

209.88.237.0/26

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CIDR and Routing

- **Aggregation** of routing table entries:

  128.143.0.0/16 and 128.142.0.0/16 can be represented as 128.142.0.0/15 at a router.

  143 = 128.10001111.0.0 142 = 128.10001110.0.0

- **Longest prefix match**: Routing table lookup finds the routing entry that matches the longest prefix

  Why????

E.g., What is the outgoing interface for destination IP address: 128.143.137.0?

Prefix	Interface/outgoing link
128.143.128.0/17	interface #1
128.128.0.0/9	interface #2
128.0.0.0/4	interface #5

Routing table

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IPv6 - IP Version 6

- **IP Version 6**
  - Is the successor to the currently used IPv4
  - Specification completed in 1994
  - Makes improvements to IPv4 (no revolutionary changes)

- One (not the only !) feature of IPv6 is a significant increase in size of the IP address to **128 bits (16 bytes)**
  - IPv6 will solve – for the foreseeable future – the problems with IP addressing

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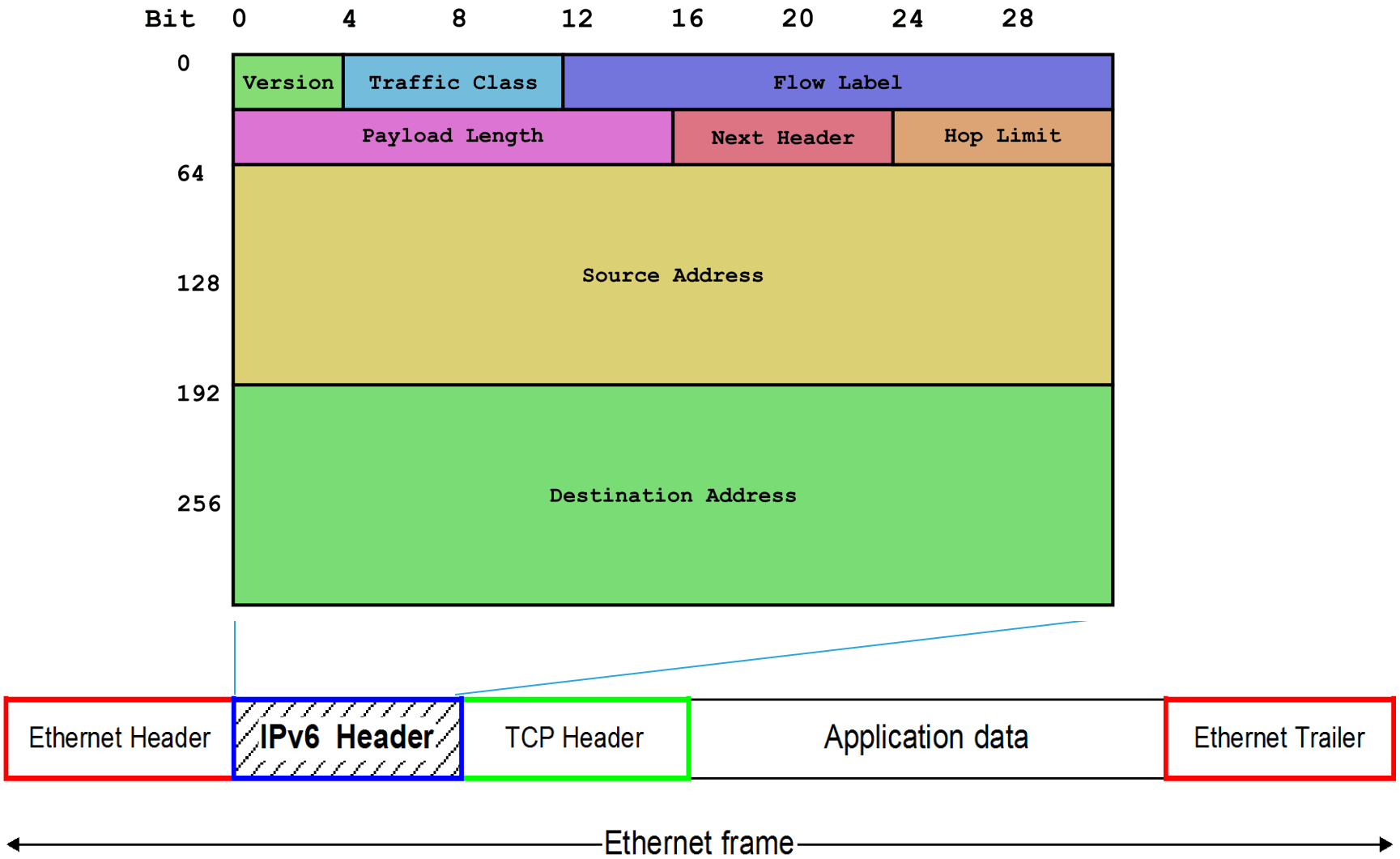
IPv6 Header



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IPv6 vs. IPv4: Address Comparison

- **IPv4** has a maximum of

  $2^{32} \approx$ 4 billion addresses

- **IPv6** has a maximum of

  $2^{128} = (2^{32})^4 \approx$ 4 billion x 4 billion x 4 billion x 4 billion addresses

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Notation of IPv6 addresses

- **Convention**: The 128-bit IPv6 address is written as **eight 16-bit integers** (using hexadecimal digits for each integer)

CEDF:BP76:3245:4464:FACE:2E50:3025:DF12

- **Short notation:**
- Abbreviations of leading zeroes:

CEDF:BP76:0000:0000:009E:0000:3025:DF12
→ CEDF:BP76:0:0:9E :0:3025:DF12

- ":0000:0000" can be written as "::"

CEDF:BP76:0:0:FACE:0:3025:DF12 → CEDF:BP76::FACE:0:3025:DF12

- IPv6 addresses derived from IPv4 addresses have different formats. Convention allows to use IPv4 notation for the last 32 bits.

128.143.137.144 -> 0:0:0:0:0:ffff:808F:8990 or

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IPv6 Provider-Based Addresses

- The first IPv6 addresses will be allocated to a provider-based plan

- Type: Set to “010” for provider-based addresses
- Registry: identifies the agency that registered the address

The following fields have a variable length (recommeded length in “()”)

- Provider: Id of Internet access provider *(16 bits)*
- Subscriber: Id of the organization at provider *(24 bits)*
- Subnetwork: Id of subnet within organization *(32 bits)*
- Interface: identifies an interface at a node *(48 bits)*

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More on IPv6 Addresses

- The provider-based addresses have a similar flavor as CIDR addresses

- IPv6 provides address formats for:

  Unicast – identifies a single interface

  Multicast – identifies a group. Datagrams sent to a multicast address are sent to all members of the group

  Anycast – identifies a group. Datagrams sent to an anycast address are sent to one of the members in the group.

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- IPv4 hanya terdiri atas 32 bit sehingga pengalamatannya sangat terbatas, dengan makin bertambahnya jumlah pengguna jaringan dibuatlah suatu metode untuk memperbanyak *Network ID* dari suatu *Network ID* yang telah ada.
- Metode ini disebut dengan istilah *Subnetting*, yaitu mengorbankan sebagian bit *Host ID* untuk digunakan membuat *Network ID* tambahan
- Perhitungan *Subnetting* meliputi 5 hal, yaitu **Subnetmask baru hasil subnetting, Jumlah subnet yang terbentuk, Jumlah host tiap subnet, Range alamat host tiap subnet, dan Alamat broadcast tiap subnet**.
- Subnetting dapat diterapkan pada alamat IP *Classful* ataupun *Classless* menggunakan notasi CIDR (*Classless Inter Domain Routing*)

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- Penulisan IP address dengan mencantumkan jumlah bit yang digunakan sebagai network ID menggunakan notasi *slash* (/), misalnya 192.168.1.2/24.
- /24 artinya sebanyak 24 bit (dari kiri) merupakan Network ID.
- Sehingga, sisa bit yang dapat digunakan untuk membuat host sebanyak 8 bit, yakni 32 – 24 = 8
- Subnetmask dalam bentuk binernya: 11111111.11111111.11111111.00000000
- Subnetmask dalam bentuk desimalnya:

  255.255.255.0.

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Subnetmask dan nilai /-nya

Subnet Mask	Nilai CIDR
255.128.0.0	/9
255.192.0.0	/10
255.224.0.0	/11
255.240.0.0	/12
255.248.0.0	/13
255.252.0.0	/14
255.254.0.0	/15
255.255.0.0	/16
255.255.128.0	/17
255.255.192.0	/18
255.255.224.0	/19

Subnet Mask	Nilai CIDR
255.255.240.0	/20
255.255.248.0	/21
255.255.252.0	/22
255.255.254.0	/23
255.255.255.0	/24
255.255.255.128	/25
255.255.255.192	/26
255.255.255.224	/27
255.255.255.240	/28
255.255.255.248	/29
255.255.255.252	/30

Dimulai dari /8 (255.0.0.0) s/d /30 (255.255.255.252) dimana setiap penambahan 1 bit untuk membuat subnet nilai / bertambah 1 dan seterusnya (Lihat tabel pada slide selanjutnya).

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Tabel IP Address dan Subnetting

IP Address								
Posisi bit	8th	7th	6th	5th	4th	3th	2nd	1st
Bobot	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Bentuk biner	10000000	01000000	00100000	00010000	00001000	00000100	00000010	00000001
Bentuk desimal	128	64	32	16	8	4	2	1

Subnetting								
Jumlah bit mask	1	2	3	4	5	6	7	8
Subnet mask	128	192	224	240	248	252	254	255
Bentuk biner	10000000	11000000	11100000	11110000	11111000	11111100	11111110	11111111
Bentuk slash	/1	/2	/3	/4	/5	/6	/7	/8
Jumlah subnet	2	4	8	16	32	64	128	256

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Contoh soal:

- Gunakanlah *Network ID* 192.168.0.0 & *Subnet Mask* 255.255.255.224.

Langkah berikutnya yang harus dilakukan adalah :

1. Dari oktet pertama *IP Address* 192.168.0.0 (“W”) dapat kita ketahui bahwa *IP Address* tersebut adalah Kelas C (karena *IP Address* tersebut berada dalam range angka 192-223)

   oktet keempat dari *Subnet Mask* diselubungkan dengan angka 224, jadi telah diambil 3 bit untuk membuat subnet.

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2. Hitung jumlah subnet yang akan terbentuk menggunakan rumus 2^n, dimana n adalah jumlah bit yang diselubungkan. $2^3 = 8$, jadi ada 8 subnet.
3. Pergunakanlah rumus (256 dikurang Angka oktet yang diselubungkan), jadi :

   256 – 224 = 32, sehingga ada 8 subnet yang terbentuk dengan kelipatan 32.
4. Maka didapatkan kelompok *subnet* baru yang dapat digunakan adalah kelipatan angka 32 dan tidak boleh melebihi angka 224, yaitu 0, 32, 64, 96, 128, 160, 192, dan 224.

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Dengan demikian maka *Subnet* baru yang terbentuk: 192.168.0.**0**, 192.168.0.**32**, 192.168.0.**64**, 192.168.0.**96**, 192.168.0.**128**, 192.168.0.**160**, 192.168.0.**192**, dan 192.168.0.**224**.

Lengkapnya adalah sebagai berikut :

Network ID : 192.168.0.0 *(Subnet-1)*
IP Address : 192.168.0.1-192.168.0.30
IP Add Broad. : 192.168.0.31
Subnet Mask : 255.255.255.224

Network ID : 192.168.0.32 *(Subnet-2)*
IP Address : 192.168.0.33-192.168.0.62
IP Add Broad. : 192.168.0.63
Subnet Mask : 255.255.255.224

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Network ID : 192.168.0.64 *(Subnet-3)*
IP Address : 192.168.0.65-192.168.0.94
IP Add Broad. : 192.168.0.95
Subnet Mask : 255.255.255.224

Network ID : 192.168.0.96 *(Subnet-4)*

IP Address : 192.168.0.97-192.168.0.126

IP Add Broad. : 192.168.0.127

Subnet Mask : 255.255.255.224

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Network ID : 192.168.0.128 *(Subnet-5)*

IP Address : 192.168.0.129-192.168.0.158

IP Add Broad. : 192.168.0.159

Subnet Mask : 255.255.255.224

Network ID : 192.168.0.160 *(Subnet-6)*

IP Address : 192.168.0.161-192.168.0.190

IP Add Broad. : 192.168.0.191

Subnet Mask : 255.255.255.224

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Network ID : 192.168.0.192 *(Subnet-7)*

IP Address : 192.168.0.193-192.168.0.222

IP Add Broad. : 192.168.0.223

Subnet Mask : 255.255.255.224

Network ID : 192.168.0.224 *(Subnet-8)*

IP Address : 192.168.0.225-192.168.0.254

IP Add Broad. : 192.168.0.255

Subnet Mask : 255.255.255.224

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Catatan Penting untuk Dipahami

1. Teknik yang digunakan untuk menghitung jumlah subnet sama dengan teknik yang digunakan untuk menghitung jumlah host, namun karena host id dalam bentuk biner tidak boleh 0 semua (network id) atau 1 semua (alamat broadcast) sedangkan untuk subnet boleh, maka rumus keduanya menjadi berbeda.
2. Rumus untuk menghitung jumlah subnet adalah 2^n
3. Rumus untuk menghitung jumlah host id yang dapat digunakan adalah $2^n – 2$, yakni dikurang 2 karena kombinasi biner 0 atau 1 semua tidak dapat digunakan.
4. Anda dapat menyelubungkan dua atau lebih bit-bit Host ID untuk mendapatkan Network ID baru selama masih tersedia bit Host ID yang dapat anda selubungkan.

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Kelas C

Network ID : 200.120.10.0/24 (Kelas C)

Subnet Mask : 255.255.255.0 (Mask Default kelas C)

Jumlah bit mask : 4 bit (N = 4)

Langkah 1 (lihat tabel Subnetting)

- Subnet Mask baru : 255.255.255.240

Langkah 2 (2^N)

- Jumlah subnet : 2^4 = 16 subnet

Langkah 3 (256 – Bobot Mask)

- Kelipatan subnet : 256 – 240 = 16

Jadi ada 16 subnet dengan kelipatan 16

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Kelas C

Langkah 4

- Daftar Subnet (nilai / awal ditambah N)

1. 200.120.10.0/28
2. 200.120.10.16/28
3. 200.120.10.32/28
4. 200.120.10.48/28
5. 200.120.10.64/28
6. 200.120.10.80/28
7. 200.120.10.96/28
8. 200.120.10.112/28
9. 200.120.10.128/28
10. 200.120.10.144/28
11. 200.120.10.160/28
12. 200.120.10.176/28
13. 200.120.10.192/28
14. 200.120.10.208/28
15. 200.120.10.224/28
16. 200.120.10.240/28

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Kelas C

Langkah 5

- Jumlah bit untuk Host ID : 32 – 28 = 4
- Maksimal Host tiap Subnet : 2^4-2
  : 16 – 2 = 14
- Jumlah maksimal seluruh Host :14 * 16
  : 224

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Kelas C

Langkah 5

- Data lengkap tiap subnet:
- Subnet ke-1

  Network ID : 200.120.10.0/28

  Range IP Address : 200.120.10.1 s/d 200.120.10.14

  Broadcast address : 200.120.10.15

  Subnetmask : 255.255.255.240
- Subnet ke-2

  Network ID : 200.120.10.16/28

  Range IP Address : 200.120.10.17 s/d 200.120.10.30

  Broadcast address : 200.120.10.31

  Subnetmask : 255.255.255.240

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Kelas C

Langkah 5

- Data lengkap tiap subnet:
- Subnet ke-3

  Network ID : 200.120.10.32/28

  Range IP Address : 200.120.10.33 s/d 200.120.10.46

  Broadcast address : 200.120.10.47

  Subnetmask : 255.255.255.240
- Subnet ke-4

  Network ID : 200.120.10.48/28

  Range IP Address : 200.120.10.49 s/d 200.120.10.62

  Broadcast address : 200.120.10.63

  Subnetmask : 255.255.255.240

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Kelas C

Langkah 5

- Data lengkap tiap subnet:
- Subnet ke-5

  Network ID : 200.120.10.64/28

  Range IP Address : 200.120.10.65 s/d 200.120.10.78

  Broadcast address : 200.120.10.79

  Subnetmask : 255.255.255.240
- Subnet ke-6

  Network ID : 200.120.10.80/28

  Range IP Address : 200.120.10.81 s/d 200.120.10.94

  Broadcast address : 200.120.10.95

  Subnetmask : 255.255.255.240

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Kelas C

Langkah 5

- Data lengkap tiap subnet:
- Subnet ke-7

  Network ID : 200.120.10.96/28

  Range IP Address : 200.120.10.97 s/d 200.120.10.110

  Broadcast address : 200.120.10.111

  Subnetmask : 255.255.255.240
- Subnet ke-8

  Network ID : 200.120.10.112/28

  Range IP Address : 200.120.10.113 s/d 200.120.10.126

  Broadcast address : 200.120.10.127

  Subnetmask : 255.255.255.240

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Kelas C

Langkah 5

- Data lengkap tiap subnet:
- Subnet ke-9

  Network ID : 200.120.10.128/28

  Range IP Address : 200.120.10.129 s/d 200.120.10.142

  Broadcast address : 200.120.10.143

  Subnetmask : 255.255.255.240
- Subnet ke-10

  Network ID : 200.120.10.144/28

  Range IP Address : 200.120.10.145 s/d 200.120.10.158

  Broadcast address : 200.120.10.159

  Subnetmask : 255.255.255.240

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Kelas C

Langkah 5

- Data lengkap tiap subnet:
- Subnet ke-11

  Network ID : 200.120.10.160/28

  Range IP Address : 200.120.10.161 s/d 200.120.10.174

  Broadcast address : 200.120.10.175

  Subnetmask : 255.255.255.240
- Subnet ke-12

  Network ID : 200.120.10.176/28

  Range IP Address : 200.120.10.177 s/d 200.120.10.190

  Broadcast address : 200.120.10.191

  Subnetmask : 255.255.255.240

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Kelas C

Langkah 5

- Data lengkap tiap subnet:
- Subnet ke-13

  Network ID : 200.120.10.192/28

  Range IP Address : 200.120.10.193 s/d 200.120.10.206

  Broadcast address : 200.120.10.207

  Subnetmask : 255.255.255.240
- Subnet ke-14

  Network ID : 200.120.10.208/28

  Range IP Address : 200.120.10.209 s/d 200.120.10.222

  Broadcast address : 200.120.10.223

  Subnetmask : 255.255.255.240

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Kelas C

Langkah 5

- Data lengkap tiap subnet:
- Subnet ke-15

  Network ID : 200.120.10.224/28

  Range IP Address : 200.120.10.225 s/d 200.120.10.238

  Broadcast address : 200.120.10.239

  Subnetmask : 255.255.255.240
- Subnet ke-16

  Network ID : 200.120.10.240/28

  Range IP Address : 200.120.10.241 s/d 200.120.10.254

  Broadcast address : 200.120.10.255

  Subnetmask : 255.255.255.240

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Kelas C

Latihan soal:

Kerjakan sampai 16 subnet pertama saja

Network ID : 200.120.10.0/24

Subnet Mask : 255.255.255.0

Jumlah bit mask :

a. 1 bit
b. 2 bit
c. 3 bit
d. 5 bit
e. 6 bit

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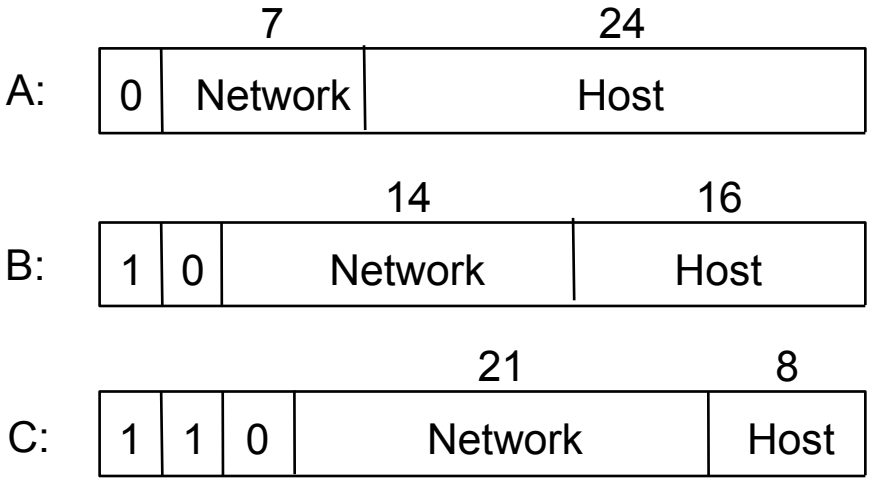
Global addressing

- Properties
  - globally unique
  - hierarchical: network + host

- Dotted Decimal Notation
  - Class A: $2^{24} - 2 = 16{,}777{,}214$ hosts
    - 1.0.0.1 – 126.255.255.254
    - (0.0.0.0 – 0.255.255.255, 1.0.0.0, 126.255.255.255, and 127.0.0.0 – 127.255.255.255 reserved)
  - Class B: $2^{16} - 2 = 65{,}534$ hosts
    - 128.0.0.1 – 191.255.255.254
  - Class C: $2^{8} - 2 = 254$ hosts
    - 192.0.0.1 – 223.255.255.254



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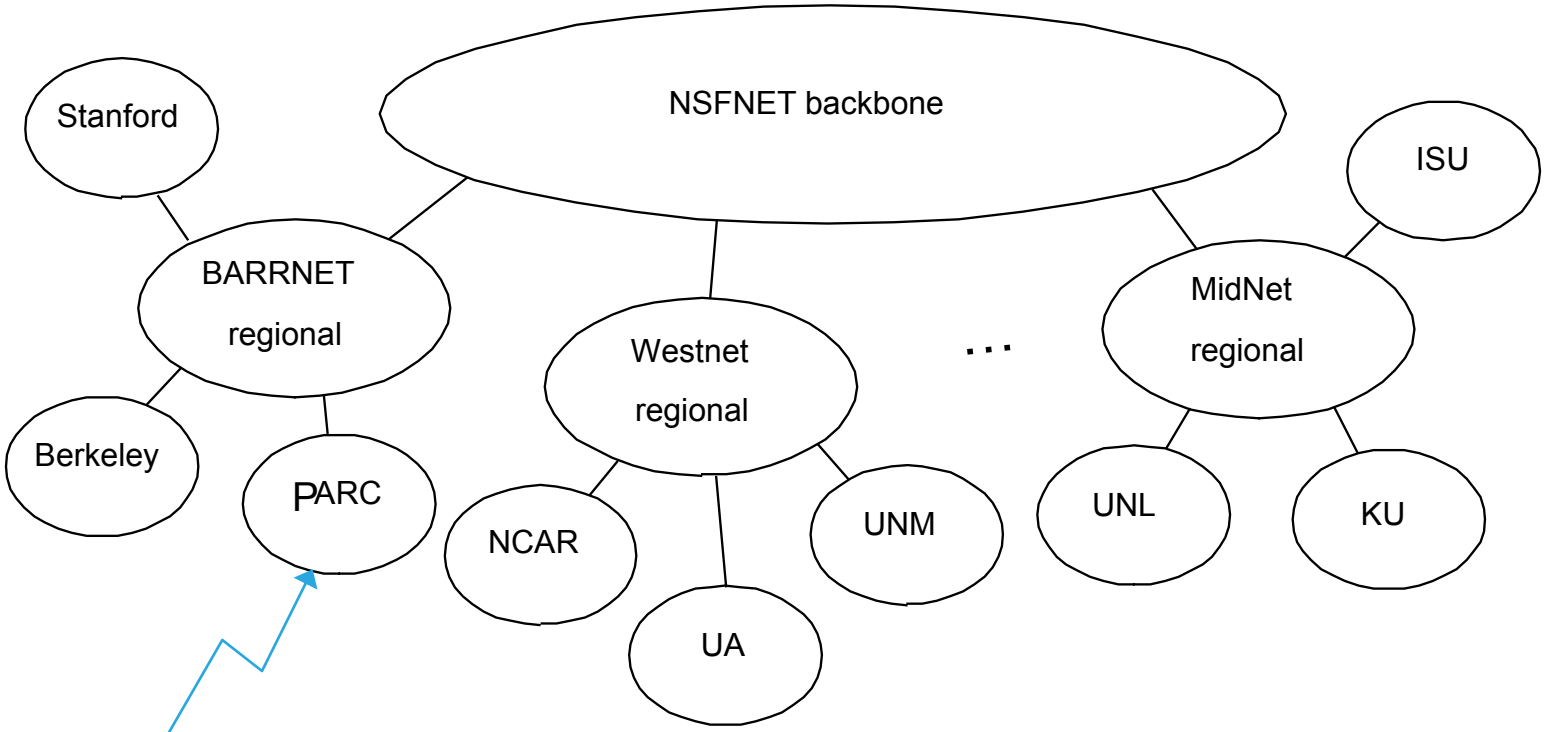
Internet Structure



- Autonomous System (AS):
  - Administered independently of other AS
  - Have a different routing protocol and metrics
- Classful Addressing: Do we really need to give an independent class A/B/C network number to every single AS?

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Scaling Issues in Routing

- Inefficient use of IP Address Space

  Class C with 2 hosts (2/254 = 0.78% efficient)

  Class B with 256 hosts (256/65534 = 0.39% efficient)

  IP address space gets consumed too quickly

- Too Many Networks

  Routing tables do not scale

  Route propagation protocols do not scale

  Router gets slower to scan a big forwarding table

Hierarchy

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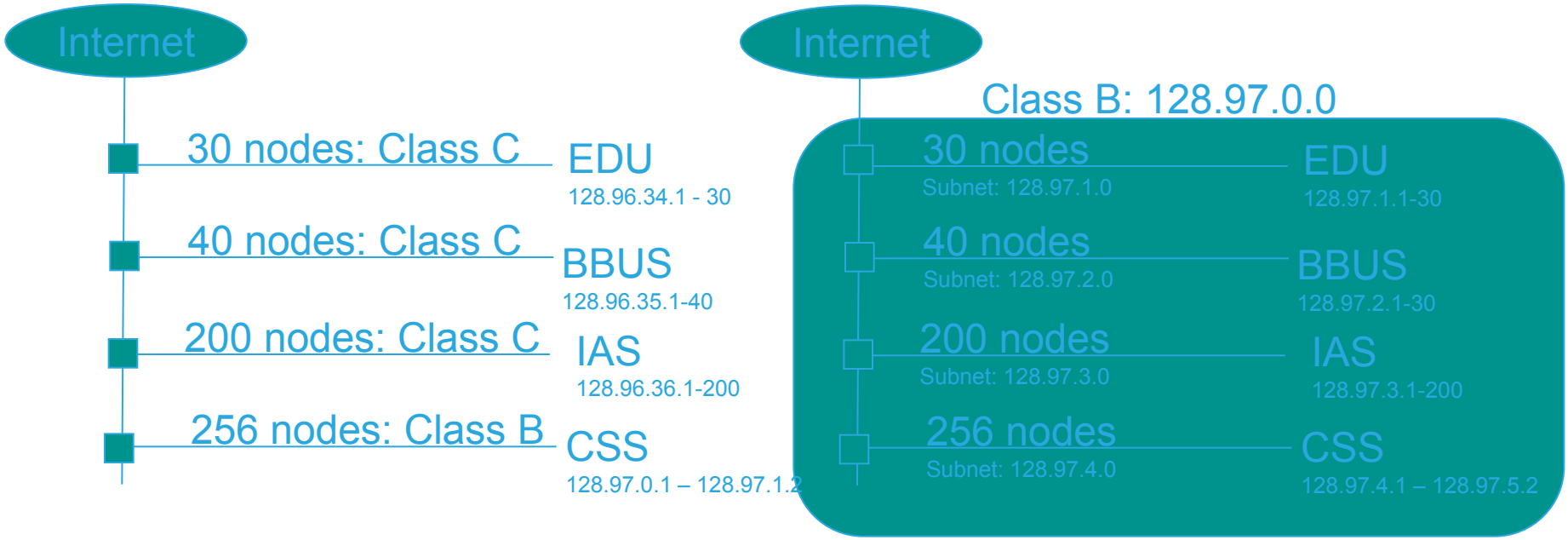
Subnetting - Concept

Simple IP networks

A collection of subnets



- Problem: Internet identifies only classes

  Four networks each must receive an independent class of network number, (which exhausts IP addresses and floods network #s)

  Subnet: collects networks belonging to the same AS and give a single class of network number, which is then divided into subnet numbers internally.

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Subnetting – How to Address

- *Subnet masks* define variable partition of host part
- Subnets visible only within site

Network number	Host number

Class B address

127.97.0.1 – 127.97.255.254

11111111111111111111111	00000000

Subnet mask (255.255.255.0)

# of bits in subnet mask

Network number	Subnet ID	Host ID

Subnetted address

127.97.8.254/24

Subnet ID

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Subnetting – How to Address

#bits	Subnetwork Mask	#subnets in Class B	#subnets in Class C	# of hosts
16	255.255.0.0	1	-	65534
17	255.255.128.0	-	-	32766
18	255.255.192.0	2	-	16382
19	255.255.224.0	6	-	8190
20	255.255.240.0	14	-	4094
21	255.255.248.0	30	-	2046
22	255.255.252.0	62	-	1022
23	255.255.254.0	126	-	510
24	255.255.255.0	254	1	254
25	255.255.255.128	510	0	126
26	255.255.255.192	1022	2	62
27	255.255.255.224	2046	6	30
28	255.255.255.240	4094	14	14
29	255.255.255.248	8190	30	6
30	255.255.255.252	16382	62	2
31	255.255.255.254	32766	126	-
32	255.255.255.255	65534	254	-

Note: subnet all 0's and all 1's are not recommended

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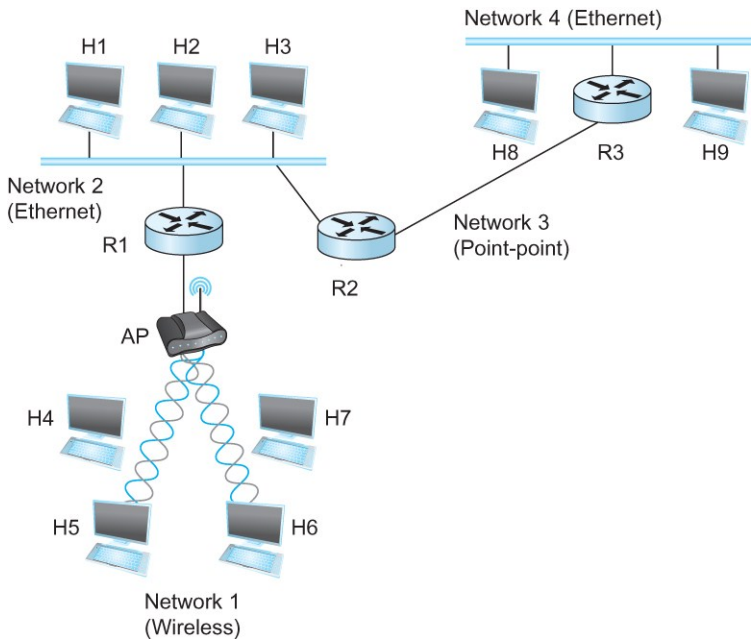
Routing with simple IP



NetworkNum	NextHop
1	R1
2	Interface 1
3	Interface 0
4	R3

[Note: NetworkNum values would typically be more like 128.96.34]

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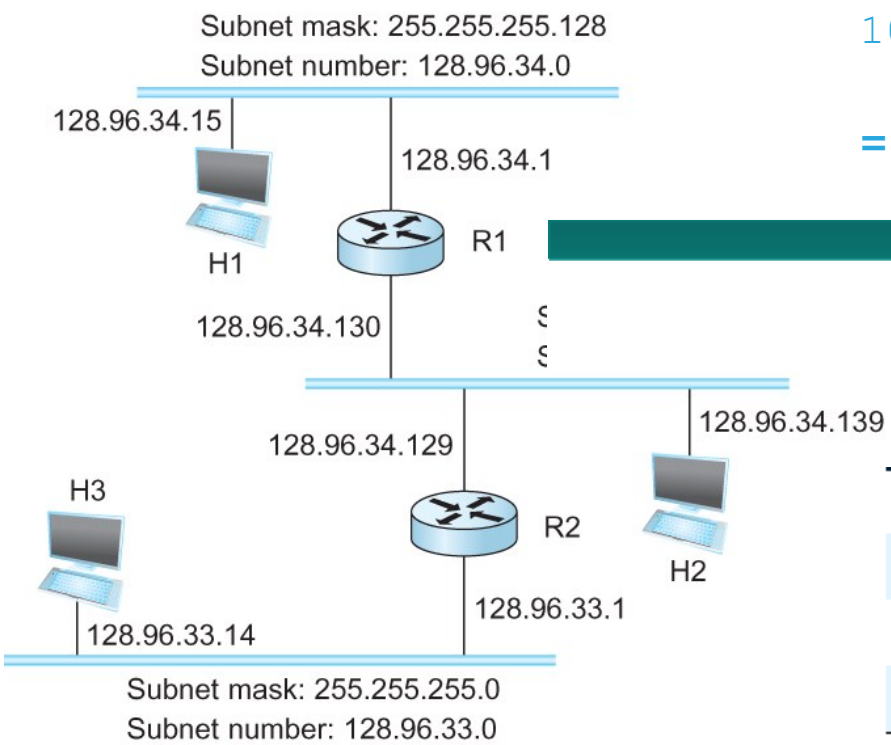
Routing with subnetting

IP address & subnet mask = subnet number
Example: 128.96.34.15 & 255.255.255.128

```
10000000.01100000.00100010.00001111 &
11111111.11111111.11111111.10000000
-----------------------------------
10000000.01100000.00100010.00000000
```

= 128.96.34.0



Forwarding Table for R1

SubnetNumber	SubnetMask	NextHop
128.96.34.0	255.255.255.128	Interface 0
128.96.34.128	255.255.255.128	Interface 1
128.96.33.0	255.255.255.0	R2

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Forwarding Algorithm

```
D = destination IP address
for each entry (SubnetNum, SubnetMask, NextHop)
   D1 = SubnetMask & D
   if D1 == SubnetNum
      if NextHop is an interface
         deliver datagram directly to destination
      else
         deliver datagram to NextHop (a router)
```

- Use a default router if nothing matches
- Not necessary for all 1s in subnet mask to be contiguous
  But highly recommended
- Can put multiple subnets on one physical network

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Supernetting

- *Sub*netting

  Purpose: divide a large class of network numbers into sub network numbers → helps assign addresses efficiently

  Problem: an AS with more than 255 hosts still needs class B

- *Super*netting

  Solution: assign block of contiguous network numbers to an institution.

  Ex. Assign two class C network numbers instead of one class B network.

  Side effect: The information that routers store and exchange increases dramatically

  Ex. If an AS has 16 class C network numbers, every Internet router needs 16 entries for this AS.

CIDR: *Classless* Inter-Domain Routing

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CIDR

- Basic concept of supernetting using class C:
  - Represent blocks with a single pair
    - `(first_class_C_network_address, count)`
  - Example: (192.5.48.0, 3)
    - Points to a sequence of blocks:
      192.5.48.0, 192.5.49.0 and 192.5.50.0
- In practice
  - No restriction to class C nor use of count
  - Restrict block sizes to powers of 2
  - Use a bit mask (CIDR mask) to identify block size
    - Ex. An AS assigned a block of 2048 (2^{11}) contiguous addresses starting at 128.211.168.0 is a collection of 8 (2^3) class C networks (with 2^8 addresses each)

Lowest	128.211.168.0	**10000000 11010011 10101**000 00000000
Highest	128.211.175.255	**10000000 11010011 10101**111 11111111
CIDR mask (32 – 11 = **21** bits)		**11111111 11111111 11111**000 00000000
Address Notation:		128.211.168.0/**21**

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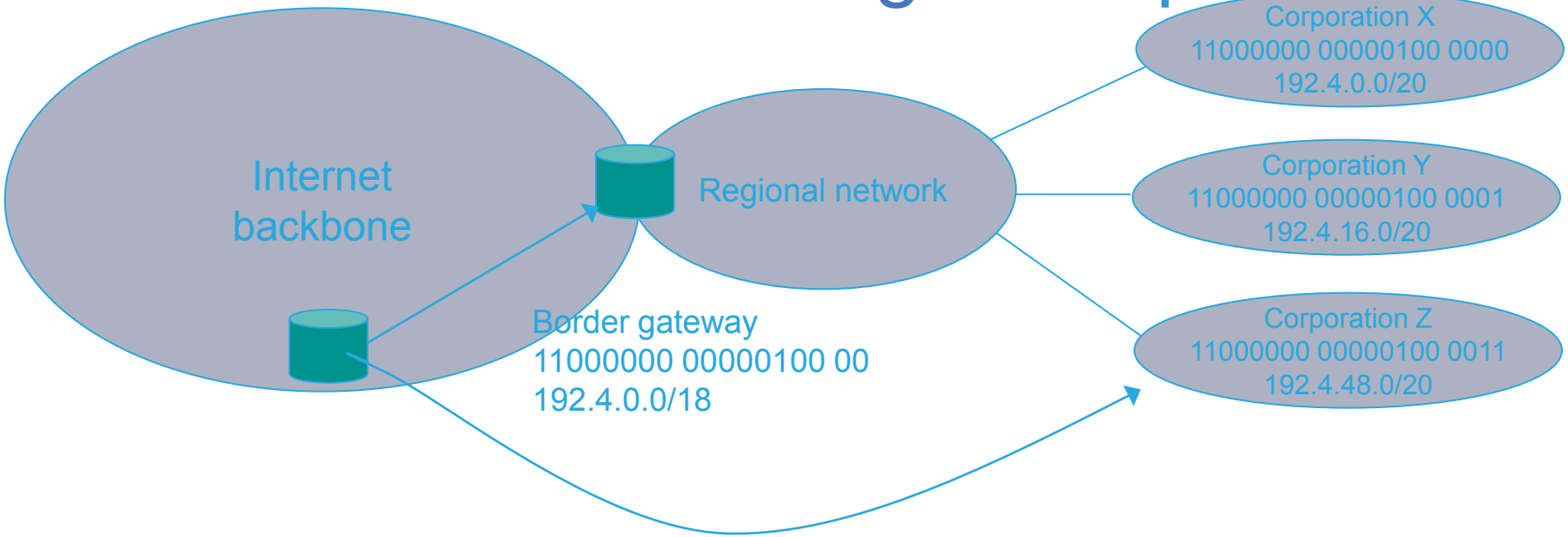
Classless Addressing Examples

- CIDR allows to aggregate routes repeatedly



- Then, what if there is a router capable of forwarding packets both to the regional network and to the corporation Z?

Prefix	Next Hop
192.4.0.0/18	the regional network
192.4.48.0/20	corporation Z

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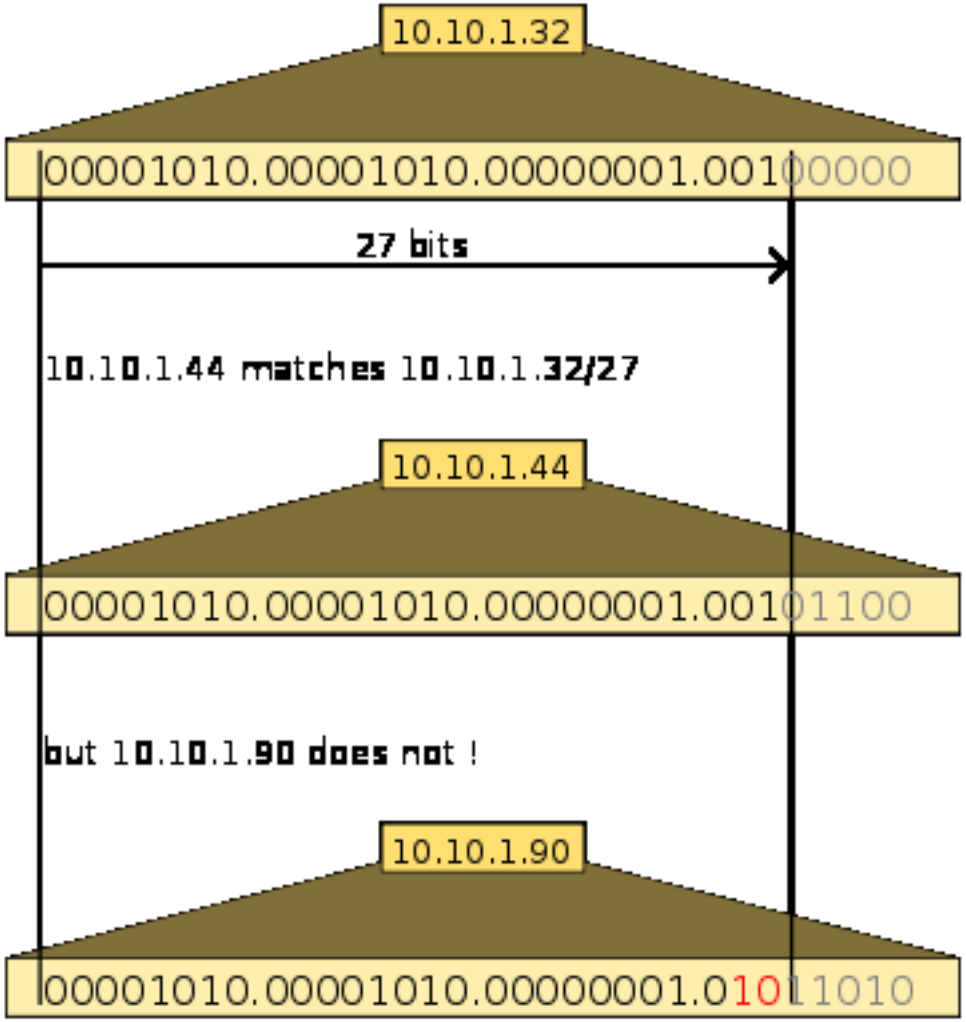

http://en.wikipedia.org/wiki/Classless_Inter-Domain_Routing

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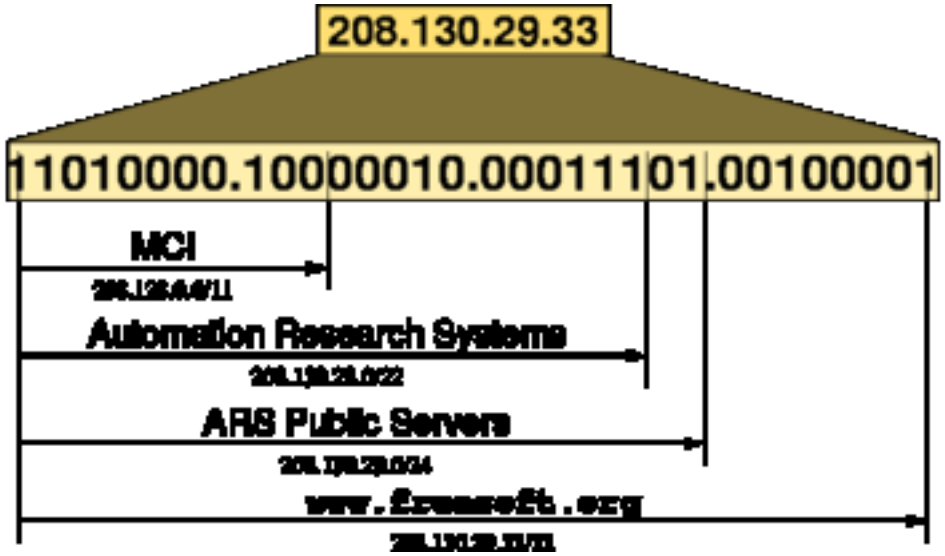

http://en.wikipedia.org/wiki/Classless_Inter-Domain_Routing

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- Reviews
  - Subnetting: How to address and forwarding algorithm
  - Supernetting: CIDR, principle of longest match, and classless lookup
- Exercises in Chapter 3
  - Ex. 55 (Subnetting)
  - Ex. 68 (CIDR)
  - Ex. 72 (CIDR)
  - Ex. 74 (CIDR)

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SOAL

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1. Mengorbankan sebagian *Host ID* untuk digunakan dalam membuat *Network ID* tambahan adalah definisi dari……

a. Network ID
b. Host ID
c. Subnetting
d. Subnet

2. Jika diambil 20 bit untuk membuat subnet, berapakah subnetmask-nya ?

a. 255.255.252.0
b. 255.255.250.0
c. 255.255.240.0
d. 255.255.224.0

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2. Jika diambil 20 bit untuk membuat subnet, berapakah subnetmask-nya ?

a. 255.255.252.0
b. 255.255.250.0
c. 255.255.240.0
d. 255.255.224.0

3. Berapa bit subnet default kelas C ?

a. 255.255.255.0
b. 24
c. 16
c. 20

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3. Berapa bit subnet default kelas C ?

a. 255.255.255.0 b. 24

c. 16 c. 20

4. Berapa jumlah subnet dari sebuah IP Address kelas C 192.168.100.0/24 yang mengalami subnetting 192.168.100.0/29 ?

a. 26 b. 28 c. 30 d. 32

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4. Berapa jumlah subnet dari sebuah IP Address kelas C 192.168.100.0/24 yang mengalami subnetting 192.168.100.0/29 ?

a. 26 b. 28 c. 30 d. 32

5. Berapa jumlah Host ID dari sebuah IP Address kelas C 192.168.100.0/24 yang mengalami subnetting 192.168.100.0/29 ?

a. 6 b. 16 c. 24 d. 32

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5. Berapa jumlah Host ID dari sebuah IP Address kelas C 192.168.100.0/24 yang mengalami subnetting 192.168.100.0/29 ?

a. 6 b. 16 c. 24 d. 32

1. Mengorbankan sebagian *Host ID* untuk digunakan dalam membuat *Network ID* tambahan adalah definisi dari……

a. Network ID b. Host ID

c. Subnetting d. Subnet

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Ex 55 (Subnetting)

- Suppose a router has built up the routing table shown in Figure 3.18. The router can deliver packets directly over interfaces 0 and 1, or it can forward packets to routers R2, R3 or R4. Describe what the router does with a packet addressed to each of the following destinations:
- (a) 128.96.39.10
- (b) 128.96.40.12
- (c) 128.96.40.151
- (d) 192.4.153.17
- (e) 192.4.153.90

SubnetNumber	SubnetMask	NextHop
128.96.39.0	255.255.255.128	Interface 0
128.96.39.128	255.255.255.128	Interface 1
128.96.40.0	255.255.255.128	R2
192.4.153.0	255.255.255.192	R3
(default)	0.0.0.0	R4

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Ex 68 (CIDR)

- An organization has been assigned the prefix 212.1.1/24 (Class C) and wants to form subnets for four departments, with hosts as follows:
  A: 75 hosts
  B: 35 hosts
  C: 20 hosts
  D: 18 hosts
  There are 148 hosts in all.
- (a) Give a possible arrangement of subnet masks to make this possible
- (b) Suggest what the organization might do if department D grows to 32 hosts

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Ex 72

- Table 3.20 is a routing table using CIDR. Address bytes are in hexadecimal. The notation “/12” in “C4.50.0.0/12” denotes a netmask with 12 leading 1 bits: FEF0.0.0. Note that the last 3 entries cover every address and thus serve in lieu of a default route. State to what next hop the following will be delivered:
- (a) C4.5E.13.87
- (b) C4.5E.22.09
- (c) C3.41.80.02
- (d) 5E.43.91.12
- (e) C4.6D.31.2E
- (f) C4.6B.31.2E

Net/MaskLength	NextHop
C4.50.0.0/12	A
C4.5E.10.0/20	B
C4.60.0.0/12	C
C4.68.0.0/14	D
80.0.0.0/1	E
40.0.0.0/2	F
00.0.0.0/2	G

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Ex 74

- An ISP that has authority to assign addresses from a /16 prefix (an old Class B address) is working with a new company to allocate it a portion of address space based on CIDR. The new company needs IP addresses for machines in 3 divisions of its corporate network: Engineering, Marketing and Sales. These divisions plan to grow as follows:
  - Engineering has 5 machines as of the start of year 1 and intends to add 1 machine every week
  - Marketing will never need more than 16 machines
  - Sales needs 1 machine for every 2 clients
- As of the start of year 1, the company has no clients, but the sales model indicates that, by the start of year 2, the company will have 6 clients and each week thereafter
  - will get one new client with probability 60%,
  - will lose one client with probability 20%, or
  - will maintain the same number with probability 20%
- (a) What address range would be required to support the company's growth plans for at least 7 years if Marketing uses all 16 of its addresses and the Sales and Engineering plans behave as expected?
- (b) How long would this address assignment last? At the time when the company runs out of address space, how would the addresses be assigned to the three groups?
- (c) If, instead of using CIDR addressing, it was necessary to use old-style classful addresses, what options would the new company have in terms of getting address space?

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